

# 6<sup>th</sup>Grid-Net: Unified Remote Sensing Image Dehazing Based on Color Restoration and Edge-Preserving

Runci Bai , Kui Jiang , Xiang Chen , Chen Wu , Dianjie Lu , Guijuan Zhang , Zhuoran Zheng

**Abstract**—Remote sensing images are frequently degraded by adverse weather conditions, particularly clouds and haze, which severely impair downstream applications. Existing restoration methods typically rely on computationally heavy architectures or sequential pipelines (*e.g.*, detail enhancement followed by color rendition) that suffer from mutual interference and artifact accumulation. Furthermore, recent unified grid-based approaches utilize fixed, isotropic interpolation kernels, neglecting the intrinsic low-dimensional manifold of natural images and inevitably causing edge blur. To address these limitations, we propose 6<sup>th</sup>Grid-Net, a highly efficient and unified remote sensing image restoration framework tailored for resource-constrained edge devices. Specifically, we construct a novel six-dimensional fusion tensor that seamlessly integrates the color rendition capabilities of 3D LUTs with the spatial-luminance detail preservation of bilateral grids. To overcome the drawbacks of standard trilinear interpolation, we introduce a manifold-adaptive high-dimensional sampling mechanism. This mechanism dynamically adjusts the interpolation kernel based on local edge orientation, texture strength, and color similarity, enabling simultaneous global color stylization and local edge refinement in a single forward pass. Additionally, an edge-aware grid smoothing constraint and dynamic quantization are incorporated to suppress ghosting artifacts and significantly compress the model size. Extensive experiments on multiple benchmark datasets demonstrate that 6<sup>th</sup>Grid-Net achieves state-of-the-art restoration quality across various degradation scenarios. Moreover, our lightweight model exhibits strong generalization in natural scene colorization and supports real-time processing of 1080P high-resolution images on consumer-grade mobile devices.

**Index Terms**—Remote Sensing Image Dehazing, 6<sup>th</sup>Grid-Net, High-Dimensional Interpolation, Edge-Preserving, Color Rendition, Lightweight Deployment.

## I. INTRODUCTION

REMOTE sensing images are crucial for surface information acquisition, underpinning applications such as agricultural assessment, urban planning, and climate change research. However, these images are frequently degraded by adverse weather conditions during acquisition, with clouds and haze being the predominant degradation factors. Clouds

Runci Bai is with the China Academy of Information and Communications Technology, Beijing 100080, China (email: 05201102@cumt.edu.cn).

Kui Jiang is with the Harbin Institute of Technology, Harbin 150000, China (email: jiangkui@hit.edu.cn).

Xiang Chen is with the Nanjing University of Science and Technology, Nanjing 210094, China (email: chenxiang@njust.edu.cn).

Chen Wu is with the National University of Defense Technology, Changsha 410073, China (email: wuchen5X@mail.ustc.edu.cn).

Dianjie Lu and Guijuan Zhang are with Shandong Normal University, Jinan 250358, China (email: {ludianjie, zhangguijuan}@sdnu.edu.cn).

Zhuoran Zheng is with Qilu University of Technology, Jinan 250353, China (email: zhengzr@njust.edu.cn).

Corresponding author: Zhuoran Zheng.

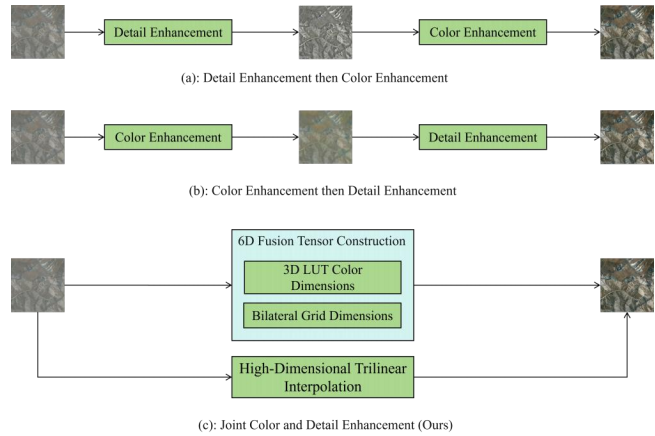


Fig. 1. Comparison of remote sensing image quality restoration paradigms: (a) Detail enhancement followed by color rendition. (b) Color rendition followed by detail enhancement. (c) The proposed 6<sup>th</sup>Grid-Net collaboratively performs color rendition and detail enhancement via a six-dimensional fusion tensor (constructed from 3D LUT color dimensions and bilateral grid spatial-luminance dimensions) and manifold-adaptive high-dimensional sampling. Our method optimizes color stylization and edge preservation simultaneously, circumventing the mutual interference and artifact accumulation inherent in sequential pipelines.

typically obscure surface information and introduce complex scattering effects, whereas haze reduces image contrast, causes color shifts, and blurs high-frequency details. The distinct physical characteristics of these two degradations pose a significant challenge for developing a unified restoration algorithm. Furthermore, the inherently high resolution of remote sensing images (*e.g.*,  $\geq 512 \times 512$ ) imposes strict real-time processing requirements on such algorithms.

Traditional remote sensing image restoration methods are typically tailored to a single degradation type, struggling to simultaneously and effectively handle both clouds and haze. Recently, “all-in-one” joint restoration methods have emerged, attempting to address multiple degradations within a unified model. By stacking deep convolutional layers, incorporating self-attention mechanisms [1]–[3], or employing state space models (SSMs) [4], [5], these approaches enhance representational capacity and achieve impressive performance on several benchmark datasets. However, the pursuit of higher restoration accuracy often entails a dramatic surge in parameter count and computational cost. This heavy computational burden renders such complex models difficult to deploy on resource-constrained edge devices, such as drones and mobile terminals, in practical application scenarios.

Beyond computational inefficiency, existing restoration pipelines face inherent architectural bottlenecks. As illustrated in Figure 1, they generally fall into two sequential

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# 6 thGrid- Net: 基于统一遥感图像去雾颜色恢复与边缘保持

Runci Bai , 蒋奎, 陈翔, 吴晨, 卢殿杰, 张桂娟, 郑卓然

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**摘要**—遥感图像经常受到恶劣天气条件的影响, 特别是云层和雾霾, 这会严重损害下游应用。现有的复原方法通常依赖于计算量大的架构或顺序流水线 (例如, 细节增强后进行色彩还原), 这些方法存在相互干扰和伪影累积的问题。此外, 最近的统一基于网格的方法利用固定的、各向同性的插值核, 忽略了自然图像固有的低维流形, 不可避免地导致边缘模糊。为了解决这些局限性, 我们提出了6<sup>th</sup>Grid-Net, 一个专为资源受限的边缘设备量身定制的高效统一遥感图像复原框架。具体来说, 我们构建了一个新颖的六维融合张量, 将3D查找表的色彩还原能力与双边网格的空间亮度细节保持无缝集成。为了克服标准三线性插值的缺点, 我们引入了一种流形自适应高维采样机制。该机制根据局部边缘方向、纹理强度和颜色相似性动态调整插值核, 从而在单次前向传播中同时实现全局色彩风格化和局部边缘细化。此外, 还引入了边缘感知网格平滑约束和动态量化, 以抑制鬼影伪影并显著压缩模型大小。在多个基准数据集上的大量实验表明, 6<sup>th</sup>Grid-Net在各种退化场景下均达到了最先进的复原质量。此外, 我们的轻量级模型在自然场景着色中表现出强大的泛化能力, 并支持在消费级移动设备上实时处理1080P高分辨率图像。

**索引术语**—遥感图像去雾、 6<sup>th</sup>Grid-Net、高维插值、边缘保持、颜色渲染、轻量化部署。

## I. 引言NTRODUCTION

遥感图像对于地表信息获取至关重要, 支撑着农业评估、城市规划和气候变化研究等应用。然而, 这些图像在获取过程中经常受到恶劣天气条件的影响而退化, 其中云层和雾霾是主要的退化因素。云层

白润辞就职于中国信息通信研究院, 北京100080, 中国 (电子邮箱: 05201102@cumt.edu.cn)。姜奎就职于哈尔滨工业大学, 哈尔滨150000, 中国 (电子邮箱: jiangkui@hit.edu.cn)。陈翔就职于南京理工大学, 南京210094, 中国 (电子邮箱: chenxiang@njust.edu.cn)。吴晨就职于国防科技大学, 长沙410073, 中国 (电子邮箱: wuchen5X@mail.ustc.edu.cn)。鲁殿杰和张桂娟就职于山东师范大学, 济南250358, 中国 (电子邮箱: {ludianjie, zhangguijuan}@sdnu.edu.cn)。郑卓然就职于齐鲁工业大学, 济南250353, 中国 (电子邮箱: zhengzr@njust.edu.cn)。通讯作者: 郑卓然。

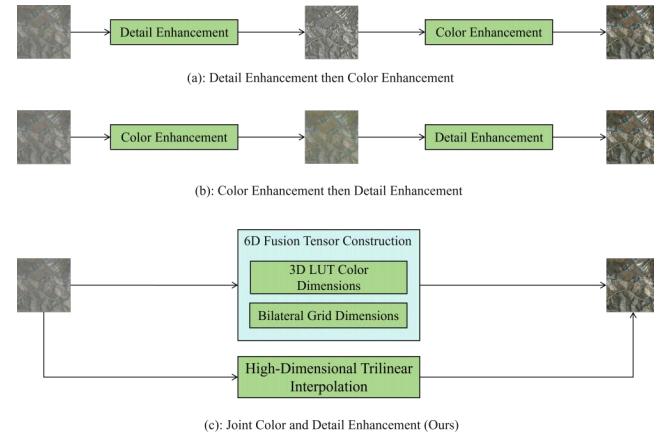


图1. 遥感图像质量恢复范式对比: (a) 先细节增强后色彩还原。(b) 先色彩还原后细节增强。(c) 所提出的 6<sup>th</sup>Grid-Net通过六维融合张量 (由三维查找表颜色维度和双边网格空间-亮度维度构建) 及流形自适应高维采样协同执行色彩还原与细节增强。我们的方法同时优化色彩风格化与边缘保持, 规避了顺序流水线中固有的相互干扰和伪影累积。

通常会遮蔽地表信息并引入复杂的散射效应, 而雾霾则会降低图像对比度、造成颜色偏移, 并模糊高频细节。这两种退化在物理特性上的显著差异, 对开发统一的复原算法构成了重大挑战。此外, 遥感图像固有的高分辨率 (例如,  $\geq 512 \times 512$ ) 对此类算法提出了严格的实时处理要求。

传统的遥感图像复原方法通常针对单一退化类型设计, 难以同时有效处理云层和雾霾。近年来, 出现了“一体化”联合复原方法, 尝试在统一模型中处理多种退化。通过堆叠深度卷积层、引入自注意力机制 [1]–[3], 或采用状态空间模型 (SSMs) [4], [5], 这些方法增强了表示能力, 并在多个基准数据集上取得了令人瞩目的性能。然而, 对更高复原精度的追求往往伴随着参数量和计算开销的急剧增加。这种沉重的计算负担使得此类复杂模型难以部署在资源受限的边缘设备上, 例如无人机和移动终端, 在实际应用场景中尤为突出。

除了计算效率低下之外, 现有的复原流程还面临着固有的架构瓶颈。如图1所示, 它们通常分为两种顺序执行的

paradigms: 1) Detail enhancement followed by color rendition (Figure 1(a)). The enhancement process may alter the original color distribution, forcing the subsequent color module to adapt to modified pixel values. Over-sharpening in this stage often leads to unnatural color transitions and local distortions. 2) Color rendition followed by detail enhancement (Figure 1(b)). The smoothing nature of color mapping attenuates original edge and texture details, complicating subsequent high-frequency restoration. Consequently, detail enhancement algorithms may excessively boost local contrast, disrupting the finely adjusted color balance and causing oversaturation or color casts.

To circumvent these sequential limitations, LUTwithBGrid [6] proposed fusing 3D LUTs with bilateral grids to incorporate spatial information in a unified manner. While promising, this method neglects the correlation between color and edge features, limiting the synergy between color mapping and edge preservation. More critically, LUTwithBGrid relies on standard trilinear interpolation, applying a fixed, isotropic kernel to all pixels regardless of local image geometry. This uniform sampling across edges, textures, and flat regions overlooks the curved manifold of natural images, inevitably introducing edge blur and ghosting artifacts.

To address these issues, we propose 6<sup>th</sup>Grid-Net, a unified and lightweight remote sensing image restoration framework. 6<sup>th</sup>Grid-Net seamlessly integrates the color rendition capability of 3D LUTs and the edge-preserving properties of bilateral grids into a novel six-dimensional fusion tensor. This architecture abandons the traditional sequential pipeline in favor of a parallel integrated design, coupling global color mapping with local detail enhancement in a single forward inference. Specifically, the first stage constructs a 6D tensor comprising color ( $R, G, B$ ) and spatial-luminance ( $X, Y, L$ ) dimensions, where each node stores enhancement values corresponding to specific spatial positions and color components. The second stage introduces a manifold-adaptive high-dimensional sampling mechanism to replace conventional trilinear interpolation. Unlike fixed-kernel interpolation, this mechanism dynamically adjusts the sampling kernel per pixel based on local edge orientation, texture strength, and color similarity. By elongating the kernel along edges to preserve continuity and suppressing it across edges to prevent blurring, it respects the intrinsic low-dimensional structure of natural images. Consequently, simultaneous color stylization and edge sharpening are achieved without multi-stage feature loss. Furthermore, we incorporate an edge-aware grid smoothing constraint and dynamic quantization to ensure robust artifact suppression and model compactness.

The proposed 6<sup>th</sup>Grid-Net is comprehensively evaluated on unified remote sensing datasets (including RICE [7] and SATEHAZE1K [8]), and its generalization capability is further validated on the ViCoW dataset [9]. It exhibits significant advantages over representative sequential methods and supports real-time processing of 1080P images on consumer-grade mobile devices. Our main **contributions** are summarized as follows:

- We propose 6<sup>th</sup>Grid-Net, a unified network that seamlessly integrates 3D LUT-based color rendition with

bilateral grid-based detail preservation into a compact six-dimensional fusion tensor. This parallel architecture enables simultaneous global color stylization and local edge refinement in a single forward pass, eliminating the computational redundancy and error accumulation inherent in sequential pipelines.

- We introduce a manifold-adaptive high-dimensional sampling mechanism that dynamically adjusts the interpolation kernel based on local edge orientation, texture strength, and color similarity. Coupled with a novel edge-aware grid smoothing constraint, it effectively suppresses noise and ghosting artifacts while preserving edge sharpness, leading to highly natural enhancement outcomes.
- To facilitate practical deployment, we employ dynamic quantization to compress the model without compromising regression accuracy, resulting in a lightweight architecture tailored for resource-constrained edge devices. Extensive experiments demonstrate that our method achieves state-of-the-art restoration quality across various adverse weather conditions and exhibits strong generalization in recovering RGB colors from grayscale natural scene images.

## II. RELATED WORK

### A. Remote sensing image restoration

Recent advancements in remote sensing image restoration can be broadly categorized into two paradigms: discriminative methods (*e.g.*, state-space models and adversarial networks) and generative methods (*e.g.*, diffusion models).

Discriminative methods typically formulate restoration as a direct mapping from degraded to clean images, leveraging structured priors or adversarial training to achieve perceptually realistic results. Within this category, state-space models (SSMs) have recently shown great promise. For instance, MambaHR [10] addresses stray light interference in hyperspectral images by integrating a state-space module with channel attention, effectively capturing global and local spectral-spatial dependencies. Similarly, DACDM-CR [11] introduces a Dynamic Cloud-Adaptive Mamba module for SAR-assisted optical data cloud removal, reconstructing global contexts with linear complexity. Adversarial training is another prominent approach. Patch-GAN [12] integrates a masked autoencoder with a patch-based discriminator to reconstruct cloud-occluded regions via transfer learning. PM-DBGAN [13] employs a physical scattering model to guide a dual-branch GAN, enabling the unsupervised inversion of atmospheric degradation for thin cloud removal. Furthermore, MAGC-GAN [14] combines multi-scale adaptive graph convolutions with a GAN architecture to capture global contextual dependencies, recovering fine details in thick cloud-covered areas.

Conversely, generative and multi-temporal methods primarily rely on iterative denoising processes or temporal feature aggregation. DBD-CR [15] adopts a dual-branch diffusion residual framework, utilizing residual learning during the diffusion process to enhance cloud removal quality. GenDeg [16] synthesizes datasets containing various degradations via a latent diffusion model, training a unified restoration network

范式: 1) 先细节增强后色彩还原 (图1(a))。增强过程可能会改变原始的色彩分布, 迫使后续的色彩模块去适应已修改的像素值。此阶段的过度锐化往往会导致不自然的色彩过渡和局部失真。2) 先色彩还原后细节增强 (图1(b))。色彩映射的平滑特性会衰减原始的边缘和纹理细节, 使后续的高频恢复变得更加困难。因此, 细节增强算法可能会过度提升局部对比度, 破坏精细调整后的色彩平衡, 从而导致过饱和或偏色。

为了规避这些顺序流水线的局限性, LUTwithBGrid [6] 提出将3D查找表与双边网格融合, 以统一的方式引入空间信息。尽管该方法具有潜力, 但它忽略了颜色特征与边缘特征之间的相关性, 限制了色彩映射与边缘保持之间的协同作用。更关键的是, LUTwithBGrid依赖标准三线性插值, 对所有像素施加固定且各向同性的核, 而不考虑局部图像几何。这种在边缘、纹理和平坦区域上不加区分的均匀采样忽视了自然图像的弯曲流形, 不可避免地引入边缘模糊和鬼影伪影。

为解决这些问题, 我们提出了 6<sup>th</sup>Grid-Net, 一个统一且轻量化的遥感图像复原框架。6<sup>th</sup>Grid-Net将3D查找表的色彩还原能力与双边网格的边缘保持特性无缝集成到一个新颖的六维融合张量中。该架构摒弃了传统的顺序流水线, 转而采用并行一体化设计, 在单次前向推理中耦合全局色彩映射与局部细节增强。具体而言, 第一阶段构建一个包含颜色 ( $R, G, B$ ) 和空间-亮度 ( $X, Y, L$ ) 维度的六维张量, 其中每个节点存储对应于特定空间位置和颜色分量的增强值。第二阶段引入流形自适应高维采样机制, 以取代传统的三线性插值。与固定核插值不同, 该机制根据局部边缘方向、纹理强度和颜色相似性, 为每个像素动态调整采样核。通过沿边缘拉长核以保持连续性, 并在跨边缘方向上抑制核以防止模糊, 它尊重了自然图像的内在低维结构。因此, 无需多阶段特征损失即可同时实现色彩风格化和边缘锐化。此外, 我们引入了边缘感知网格平滑约束和动态量化, 以确保稳健的伪影抑制和模型紧凑性。

所提出的 6<sup>th</sup>Grid-Net在统一遥感数据集 (包括 RICE [7] 和SATEHAZE1K [8]) 上进行了全面评估, 其泛化能力进一步在ViCoW数据集 [9]上得到验证。与代表性的串行方法相比, 它展现出显著优势, 并支持在消费级移动设备上对1080P图像进行实时处理。我们的主要贡献总结如下:

- 我们提出了6<sup>th</sup>Grid-Net, 一个统一网络, 将基于三维查找表的色彩还原与

基于双边网格的细节保持无缝集成到一个紧凑的六维融合张量中。这种并行架构能够在单次前向传播中同时实现全局色彩风格化和局部边缘细化, 消除了顺序流水线中固有的计算冗余和误差累积。

- 我们引入了一种流形自适应高维采样机制, 该机制根据局部边缘方向、纹理强度和颜色相似性动态调整插值核。结合一种新颖的边缘感知网格平滑约束, 它在保持边缘锐利度的同时有效抑制噪声和鬼影伪影, 从而获得高度自然的增强效果。

- 为便于实际部署, 我们采用动态量化来压缩模型, 同时不影响回归精度, 从而形成适合资源受限边缘设备的轻量级架构。大量实验表明, 我们的方法在多种恶劣天气条件下实现了最先进的复原质量, 并在从灰度自然场景图像恢复RGB颜色方面表现出强大的泛化能力。

## 二、RELATED WORK

### A. 遥感图像复原

近期遥感图像复原的进展大致可分为两类范式: 判别式方法 (如状态空间模型和对抗网络) 和生成式方法 (如扩散模型)。

判别式方法通常将复原建模为从退化图像到清晰图像的直接映射, 利用结构化先验或对抗训练来获得感知上逼真的结果。在此类别中, 状态空间模型 (SSM) 近来展现出巨大潜力。例如, MambaHR [10] 通过将状态空间模块与通道注意力相结合来解决高光谱图像中的杂散光干扰问题, 有效捕获全局和局部光谱-空间依赖性。类似地, DACDM-CR [11]为SAR辅助光学数据云去除引入了动态云自适应Mamba模块, 以线性复杂度重建全局上下文。对抗训练是另一种重要方法。Patch-GAN [12] 将掩码自编码器与基于图像块的判别器相结合, 通过迁移学习重建被云遮挡的区域。PM-DBGAN [13] 采用物理散射模型来引导双分支GAN, 实现大气退化的无监督反演, 用于薄云去除。此外, MAGC-GAN [14]将多尺度自适应图卷积与GAN架构相结合, 以捕获全局上下文依赖性, 恢复厚云覆盖区域的精细细节。

相反, 生成式与多时相方法主要依赖迭代去噪过程或时序特征聚合。DBD-CR [15] 采用双分支扩散残差框架, 在扩散过程中利用残差学习来提升云去除质量。GenDeg [16] 通过潜在扩散模型合成包含多种退化的数据集, 训练一个具有强泛化能力的统一复原网络

with strong generalization. To address thick clouds, IRNet [17] employs an implicit reconstruction approach using multi-temporal information without relying on explicit cloud masks. SGDM [18] leverages a pre-trained generative prior and integrates structural semantics from vector maps to achieve large-factor super-resolution, mitigating semantic inaccuracy and texture blurring. GMDiT [19] extends the masked DiT architecture by capturing spatial and semantic relationships through graph prior modeling, thereby enhancing structural fidelity.

Despite these advances, inherent limitations persist. GAN-based approaches frequently suffer from training instability and mode collapse, compromising performance consistency across diverse cloud and haze conditions. Diffusion-based and multi-temporal methods, while achieving high fidelity, demand massive computational resources and prolonged inference times, precluding real-time or on-orbit deployment. Additionally, multi-temporal frameworks heavily depend on cloud-free auxiliary images—often unavailable in persistently cloudy regions. Most critically, many existing methods treat cloud removal as a pure mathematical reconstruction task, neglecting the physical atmospheric scattering processes and intrinsic spectral characteristics, which inevitably leads to spectral distortion in downstream applications.

### B. Unified frameworks for multi-degradation restoration

To address the “all-in-one” image restoration task, existing unified frameworks generally fall into three strategies: degradation-aware prompting, high-capacity network architectures, and generative modeling.

The first strategy explicitly models degradation types using contrastive learning or prompt mechanisms. AirNet [20] introduced contrastive learning with a degradation encoder to map different weather corruptions into separate feature spaces. Perceive-IR [21] employs a fine-grained quality perceiver for multi-level prompt learning, enabling superior perception of degradation severity. Yang et al. [22] utilize natural language descriptions as conditional prompts for unified processing, while Wu et al. [23] propose a lightweight prompt-based framework that dynamically guides reconstruction through degradation- and instance-level prompts, mitigating catastrophic forgetting. The second strategy employs powerful backbones to uniformly handle multiple degradations via multi-scale feature fusion or kernel modulation. TransWeather [24] adopts a Transformer-based single encoder-decoder architecture with window-based self-attention to simultaneously remove rain, fog, and snow. Cui et al. [25] propose a multi-scale kernel modulation module (comprising global, large-kernel, and local branches) to efficiently learn universal representations. The third strategy exploits generative models, offering strong theoretical unification by treating various degradations as the reverse process of a common generation procedure, as demonstrated by WeatherDiffusion [26] through iterative denoising.

Recently, emerging studies have begun to prioritize real-time efficiency for deployment on resource-constrained edge devices. RTEE-Net [27] integrates global brightness and local

details via a holographic attention mechanism, achieving low-light enhancement in just 0.002 seconds with merely 50K parameters. LLF-LUT++ [28] combines Laplacian pyramid decomposition with image-adaptive 3D LUTs to deliver 4K image enhancement in 13 milliseconds.

Although these unified frameworks demonstrate strong multi-degradation capabilities, they present several drawbacks. Methods relying on predefined degradation priors often struggle to generalize to unseen complex weather conditions. Meanwhile, frameworks utilizing heavy backbones or iterative generative models incur prohibitive computational costs, rendering them unsuitable for resource-constrained platforms. While some recent methods achieve real-time processing, they often compromise restoration quality when faced with severe degradations. Most importantly, existing unified frameworks typically entangle color restoration with detail enhancement without explicitly modeling their interaction, leading to sub-optimal trade-offs between color fidelity and structural sharpness—a critical bottleneck that our proposed method aims to resolve.

## III. METHODOLOGY

### A. Overview

We propose 6<sup>th</sup>Grid-Net, as illustrated in Figure 2, an end-to-end method for remote sensing image dehazing that integrates global color mapping and local detail enhancement into a six-dimensional tensor, enabling collaborative optimization via a single high-dimensional interpolation. Given a hazy input image  $\mathbf{I}$ , it is first converted to linear RGB space, and its luminance map  $\mathbf{LM}$  is computed. In the first stage, a lightweight U-Net predicts a weight grid  $\mathbf{B}$ , which is then used to construct a fusion tensor  $\mathcal{T}$  that blends a pre-defined 3D LUT with original pixel information. In the second stage, each pixel is processed through manifold-adaptive high-dimensional sampling over this tensor to obtain the enhanced result.

### B. Bilateral Grid Weights Generation

Given the hazy input image  $\mathbf{I}_{\text{in}} \in \mathbb{R}^{3 \times H \times W}$  and its luminance map  $\mathbf{L} \in \mathbb{R}^{H \times W}$ , we aim to predict a bilateral grid weight  $\mathbf{W} \in [0, 1]^{N_x \times N_y \times N_l}$  that controls the contribution of local details during fusion. Here,  $N_x$  and  $N_y$  are the spatial grid resolutions (set to  $128 \times 72$  for 4K images), and  $N_l$  is the number of luminance bins ( $N_l = 24$ ).

As illustrated in the left branch of Figure 2, we employ a lightweight U-Net to generate  $\mathbf{W}$ . The U-Net consists of an encoder that progressively downsamples the input to capture multi-scale features, and a decoder that upsamples the features back to the target grid size. Skip connections concatenate encoder features with decoder features at corresponding scales, preserving spatial details. The final output layer uses a  $3 \times 3$  convolution followed by a sigmoid activation to ensure the weights are bounded in  $[0, 1]$ .

Formally, the U-Net maps the input image to the grid weight:

$$\mathbf{W} = \text{UNet}(\mathbf{I}_{\text{in}}), \quad (1)$$

where each entry  $u_{i,j,k} \in [0, 1]$  represents the degree of local detail preservation at spatial location  $(i, j)$  and luminance bin

为应对厚云, IRNet[17] 采用隐式重建方法, 利用多时相信息而不依赖显式云掩膜。SGDM [18] 利用预训练的生成先验, 并融合矢量地图中的结构语义来实现大倍率超分辨率, 从而缓解语义不准确性和纹理模糊问题。GMDiT [19] 通过图先验建模捕捉空间与语义关系, 扩展了掩码 DiT 架构, 进而增强结构保真度。

尽管取得了这些进展, 固有的局限性依然存在。基于生成对抗网络的方法经常遭受训练不稳定和模式崩溃的困扰, 影响了在各种云雾条件下的性能一致性。基于扩散模型和多时相的方法虽然能够实现高保真度, 但需要大量的计算资源和较长的推理时间, 这阻碍了实时处理或在轨部署。此外, 多时相框架严重依赖无云辅助影像——而这些影像在持续多云地区往往难以获得。最关键的是, 许多现有方法将云去除视为纯粹的数学重建任务, 忽视了大气的物理散射过程和内在光谱特征, 这不可避免地导致下游应用中的光谱失真。

### B. 多退化恢复的统一框架

为了应对“一体化”图像复原任务, 现有的统一框架通常可归为三种策略: 退化感知提示、大容量网络架构和生成式建模。

第一种策略利用对比学习或提示机制显式建模退化类型。AirNet [20]引入了带有退化编码器的对比学习, 将不同的天气损坏映射到分离的特征空间中。Perceive-IR [21] 采用细粒度质量感知器进行多级提示学习, 从而实现退化严重程度的优越感知。Yang等人 [22] 利用自然语言描述作为统一处理的条件提示, 而Wu等人[23] 提出了一种轻量级基于提示的框架, 通过退化级和实例级提示动态引导重建, 缓解灾难性遗忘。第二种策略采用强大的骨干网络, 通过多尺度特征融合或核调制统一处理多种退化。TransWeather[24] 采用基于Transformer的单编码器-解码器架构, 并结合基于窗口的自注意力, 同时去除雨、雾和雪。Cui等人 [25] 提出了一种多尺度核调制模块 (包含全局、大核和局部分支), 以高效学习通用表示。第三种策略利用生成式模型, 通过将各种退化视为共同生成过程的逆向过程, 提供强大的理论统一性, 正如WeatherDiffusion [26] 通过迭代去噪所展示的那样。

LLF-LUT++ [28] 将拉普拉斯金字塔分解与图像自适应 3D LUT相结合, 在13毫秒内实现4K图像增强。

尽管这些统一框架展现出强大的多重退化处理能力, 但它们也存在若干不足。依赖预定义退化先验的方法往往难以泛化到未见过的复杂天气条件。同时, 使用重型骨干网络或迭代式生成模型的框架会产生高昂的计算成本, 使其不适合资源受限的平台。虽然一些近期方法实现了实时处理, 但在面对严重退化时往往会牺牲复原质量。最重要的是, 现有统一框架通常将颜色恢复与细节增强纠缠在一起, 而未明确建模二者之间的交互, 导致色彩保真度与结构清晰度之间的权衡欠佳——这是本方法旨在解决的关键瓶颈。

## III. METHODOLOGY

### A. 概述

我们提出 6<sup>th</sup>Grid-Net, 如图2所示, 这是一种端到端的遥感图像去雾方法, 将全局色彩映射与局部细节增强整合到一个六维张量中, 通过单次高维插值实现协同优化。给定有雾输入图像 $\mathbf{I}$ , 首先将其转换到线性 RGB 空间, 并计算其亮度图 $\mathbf{LM}$ 。在第一阶段, 轻量级 U-Net 预测权重网格 $\mathbf{B}$ , 然后利用该网格构建融合张量 $\mathcal{T}$ , 将预定义的三维查找表与原始像素信息融合。在第二阶段, 每个像素通过在该张量上进行流形自适应高维采样, 得到增强结果。

### B. 双边网格权重生成

给定有雾输入图像 $\mathbf{I}_{\text{in}} \in \mathbb{R}^{3 \times H \times W}$  及其亮度图 $\mathbf{L} \in \mathbb{R}^{H \times W}$ , 我们的目标是预测一个双边网格权重 $\mathbf{W} \in [0, 1]^{N_x \times N_y \times N_l}$ , 用于在融合过程中控制局部细节的贡献。其中,  $N_x$  和  $N_y$  是空间网格分辨率 (对于4K图像设置为  $128 \times 72$ ),  $N_l$  是亮度分箱的数量 ( $N_l = 24$ )。

如图2左侧分支所示, 我们采用一个轻量级U-Net来生成 $\mathbf{W}$ 。该U-Net由一个编码器和一个解码器组成, 编码器逐步对输入进行下采样以捕获多尺度特征, 解码器则将特征上采样回目标网格大小。跳跃连接在相应尺度上将编码器特征与解码器特征进行拼接, 从而保留空间细节。最终输出层使用一个  $3 \times 3$ 卷积, 其后接sigmoid激活, 以确保权重被限制在  $[0, 1]$ 范围内。

形式上, U-Net将输入图像映射为网格权重:

$$\mathbf{W} = \text{UNet}(\mathbf{I}_{\text{in}}), \quad (1)$$

其中每个条目 $u_{i,j,k} \in [0, 1]$  表示空间位置 $(i,j)$ 和亮度分箱

近年来, 新兴研究开始优先考虑在资源受限的边缘设备上部署时的实时效率。RTEE-Net [27]通过全息注意力机制融合全局亮度与局部细节, 以仅50K参数量在0.002秒内实现低光增强。

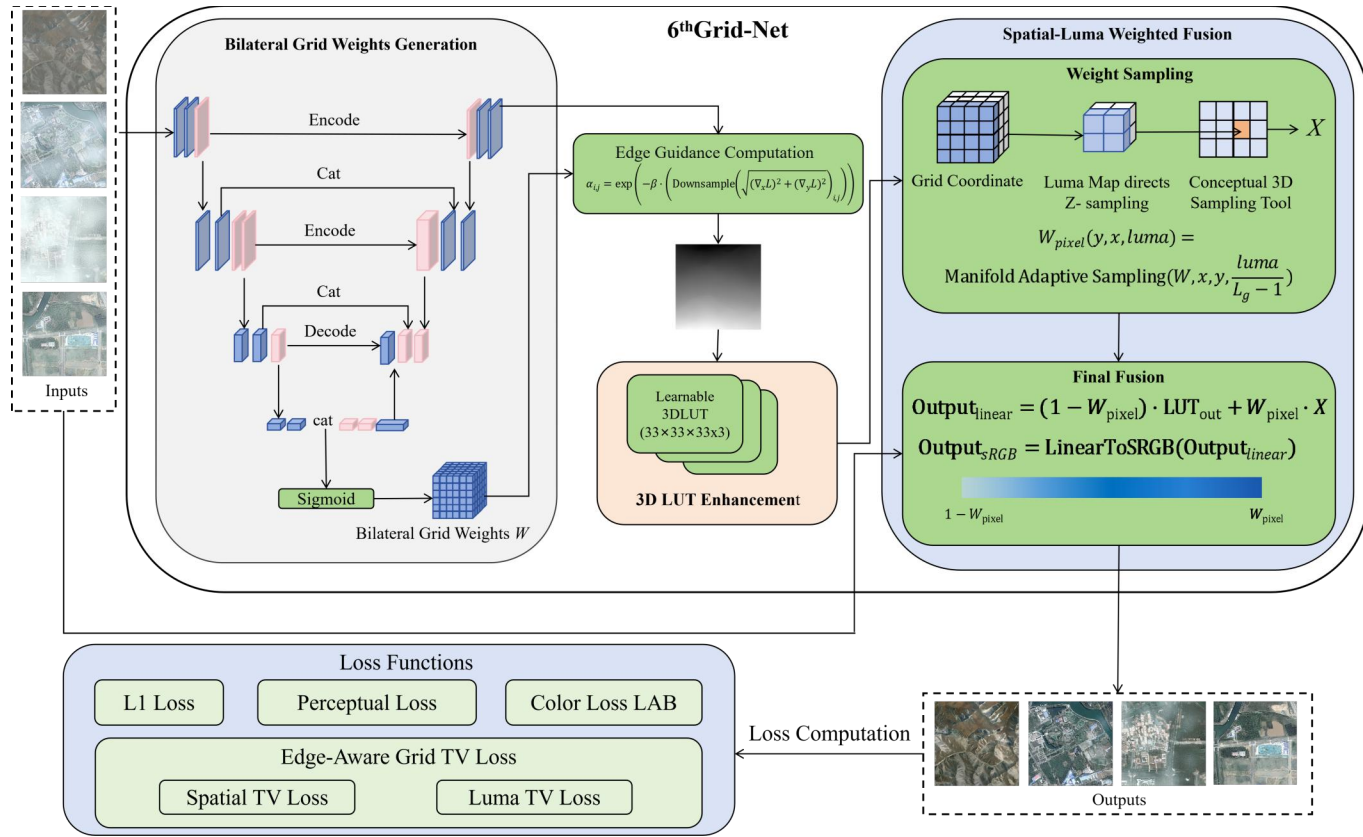


Fig. 2. Overall architecture of the proposed 6thGrid-Net, featuring a bilateral weight generation module and a spatial-luma weighted fusion mechanism. The bilateral weight grid is produced by an encoder-decoder network with skip connections, guided by an edge-aware term derived from the luminance map. Pixel-wise weights are then sampled via manifold-adaptive high-dimensional sampling based on spatial coordinates, luminance values, and local edge orientation, enabling adaptive fusion of the LUT-enhanced result with the original image. The model is optimized with a composite loss comprising L1, perceptual, color, and edge-aware TV terms.

$k$ . A value of  $w_{i,j,k} \approx 1$  indicates that the original pixel details are preserved, while  $w_{i,j,k} \approx 0$  favors the global color mapping from the 3D LUT. This adaptive mechanism allows the model to selectively retain fine structures in textured regions while ensuring color consistency in smooth areas.

### C. Edge-Aware Smoothing Regularization

Directly learning the bilateral grid weight  $\mathbf{W}$  from data may lead to undesirable local noise or blocky artifacts, as the network might exploit high-frequency variations to overfit training samples. To address this issue, we introduce an edge-aware total variation (TV) regularization that encourages  $\mathbf{W}$  to vary smoothly in homogeneous regions while allowing sharp transitions near image edges.

We first compute the gradient magnitude of the input luminance map  $\mathbf{L}$  to indicate edge strength. Using Sobel operators  $\nabla_x$  and  $\nabla_y$ , the gradient magnitude  $\mathbf{G} \in \mathbb{R}^{H \times W}$  is obtained as:

$$G = \sqrt{(\nabla_x \mathbf{L})^2 + (\nabla_y \mathbf{L})^2}. \quad (2)$$

Since the grid weight  $\mathbf{W}$  has a lower spatial resolution  $N_x \times N_y$ , we downsample  $\mathbf{G}$  to match the grid dimensions via bilinear interpolation, yielding  $\mathbf{G}_{\text{grid}} \in \mathbb{R}^{N_x \times N_y}$ . The result is then replicated across the  $N_l$  luminance bins to form  $\mathbf{G}_{\text{grid}} \in \mathbb{R}^{N_x \times N_y \times N_l}$ , as edges are independent of the luminance layer.

To attenuate regularization at edges, we define an edge-aware weighting factor:

$$\alpha_{i,j,k} = \exp(-\beta \cdot \mathbf{G}_{\text{grid}}(i, j, k)), \quad (3)$$

where  $\beta > 0$  controls the decay rate. In flat regions ( $\mathbf{G}_{\text{grid}} \approx 0$ ),  $\alpha \approx 1$  imposes strong smoothness; near strong edges ( $\mathbf{G}_{\text{grid}}$  large),  $\alpha \approx 0$  relaxes the penalty, allowing weight changes to preserve edge sharpness.

The spatial TV term penalizes differences between neighboring grid nodes in the spatial dimensions:

$$R_{\text{spatial}} = \sum_{i,j,k} \alpha_{i,j,k} (|W_{i+1,j,k} - W_{i,j,k}| + |W_{i,j+1,k} - W_{i,j,k}|), \quad (4)$$

where the sums run over valid indices. This term is weighted by  $\alpha$  to encourage smoothness in flat areas while preserving edges.

Additionally, we enforce smoothness along the luminance dimension to avoid abrupt transitions between adjacent luminance layers:

$$R_{\text{luma}} = \sum_{i,j,k} |W_{i,j,k+1} - W_{i,j,k}|, \quad (5)$$

with  $k < N_l - 1$ . No edge guidance is needed here, as luminance variations are inherently smooth.

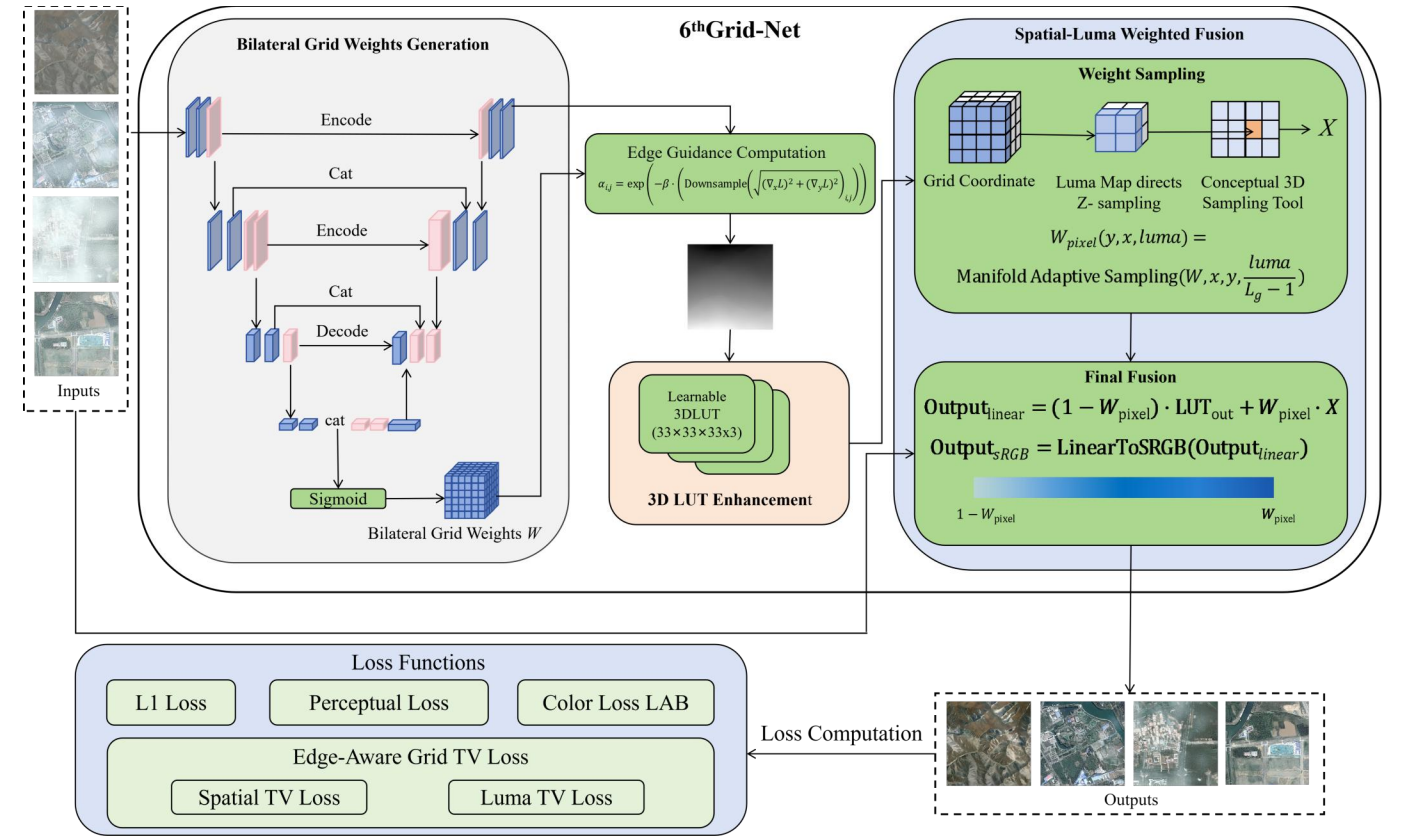


图2. 所提出的 6thGrid-Net 的总体架构，包括双边权重生成模块和空间-亮度加权融合机制。双边权重网格由带跳跃连接的编码器-解码器网络生成，并由源自亮度图的边缘感知项引导。然后，基于空间坐标、亮度值和局部边缘方向，通过流形自适应高维采样获取逐像素权重，从而能够自适应地将LUT增强结果与原始图像融合。该模型采用复合损失进行优化，包括L1、感知损失、颜色损失和边缘感知TV项。

$k$ 处的局部细节保留程度。 $w_{i,j,k} \approx 1$  的值表示保留原始像素细节，而  $w_{i,j,k} \approx 0$  则倾向于采用三维查找表的全局色彩映射。这种自适应机制使模型能够在纹理区域有选择地保留精细结构，同时在平滑区域确保颜色一致性。

### C. 边缘感知平滑正则化

直接从数据中学习双边网格权重  $\mathbf{W}$  可能会导致不良的局部噪声或块状伪影，因为网络可能利用高频变化来拟合训练样本。为解决这一问题，我们引入了一种边缘感知全变分 (TV) 正则化，鼓励  $\mathbf{W}$  在均匀区域内平滑变化，同时允许在图像边缘附近出现锐利过渡。

我们首先计算输入亮度图  $\mathbf{L}$  的梯度幅值来表示边缘强度。利用 Sobel 算子  $\nabla_x$  和  $\nabla_y$ ，得到梯度幅值  $\mathbf{G} \in \mathbb{R}^{H \times W}$  如下：

$$G = \sqrt{(\nabla_x \mathbf{L})^2 + (\nabla_y \mathbf{L})^2}. \quad (2)$$

由于网格权重  $\mathbf{W}$  具有较低的空间分辨率  $N_x \times N_y$ ，我们通过双线性插值对  $\mathbf{G}$  进行下采样以匹配网格尺寸，得到  $\mathbf{G}_{\text{grid}} \in \mathbb{R}^{N_x \times N_y}$ 。然后将其在  $N_l$  个亮度分箱上复制，形成  $\mathbf{G}_{\text{grid}} \in \mathbb{R}^{N_x \times N_y \times N_l}$ ，因为边缘与亮度层无关。

为了在边缘处减弱正则化，我们定义了一个边缘感知加权因子：

$$\alpha_{i,j,k} = \exp(-\beta \cdot \mathbf{G}_{\text{grid}}(i, j, k)), \quad (3)$$

其中  $\beta > 0$  控制衰减率。在平坦区域 ( $\mathbf{G}_{\text{grid}} \approx 0$ )， $\alpha \approx 1$  施加较强的平滑约束；在强边缘附近 ( $\mathbf{G}_{\text{grid}}$  较大)， $\alpha \approx 0$  放松惩罚，允许权重变化以保持边缘锐度。

空间TV项惩罚空间维度上相邻网格节点之间的差异：

$$R_{\text{spatial}} = \sum_{i,j,k} \alpha_{i,j,k} (|W_{i+1,j,k} - W_{i,j,k}| + |W_{i,j+1,k} - W_{i,j,k}|), \quad (4)$$

其中求和遍历所有有效索引。该项由  $\alpha$  加权，以在平坦区域促进平滑性，同时保留边缘。

此外，我们在亮度维度上施加平滑性，以避免相邻亮度层之间出现突变过渡：

$$R_{\text{luma}} = \sum_{i,j,k} |W_{i,j,k+1} - W_{i,j,k}|, \quad (5)$$

其中  $k < N_l - 1$ 。此处无需边缘引导，因为亮度变化本身是平滑的。

The overall regularization loss combines both terms:

$$L_{TV} = \lambda_s R_{\text{spatial}} + \lambda_l R_{\text{luma}}, \quad (6)$$

where  $\lambda_s$  and  $\lambda_l$  are scalar weights (set to 0.01 in practice). By incorporating  $L_{TV}$  into the training objective, the network learns to produce weight grids that are spatially coherent in smooth regions yet adaptive to image structures, effectively suppressing artifacts while maintaining sharp details.

#### D. Construction of Six-Dimensional Fusion Tensor

Having obtained the bilateral weight grid  $W$  and the learnable 3D LUT  $LUT \in \mathbb{R}^{N_c \times N_c \times N_c \times 3}$ , we now construct a six-dimensional fusion tensor  $T$  that jointly encodes global color mapping and spatially varying detail preservation. The tensor  $T$  is defined over six dimensions: spatial coordinates  $X$  and  $Y$ , luminance level  $L$ , and RGB color indices  $R, G, B$ . Its shape is  $\mathbb{R}^{N_x \times N_y \times N_l \times N_c \times N_c \times N_c \times 3}$ , where  $N_x, N_y$  denote the spatial grid resolutions,  $N_l$  the number of luminance bins, and  $N_c$  the 3D LUT size. The final dimension stores the output linear RGB value for each combination of indices.

To balance computational efficiency and restoration quality, we set  $N_x = 128$ ,  $N_y = 72$ ,  $N_l = 24$ , and  $N_c = 33$  for high-resolution remote sensing images. The spatial downsampling factor (approximately  $30\times$ ) reduces the tensor size to about 1.935 million parameters, while the luminance dimension and LUT size provide sufficient granularity for accurate color styling and detail adaptation.

The tensor is populated by blending two information sources at each grid node: the global color mapping from the 3D LUT and the original pixel values from the input image. For a given grid node with spatial indices  $(i, j)$  and luminance index  $k$ , we first sample the linear RGB image  $I_{lin}$  at the center of the corresponding grid cell, obtaining  $I_{\text{sample}}(i, j) \in \mathbb{R}^3$ . The bilateral weight  $w_{i,j,k} \in [0, 1]$  from  $W$  controls the contribution of local details. The tensor entry for color indices  $(r, g, b)$  is then defined as:

$$T(i, j, k, r, g, b) = (1 - w_{i,j,k}) \cdot LUT(r, g, b) + w_{i,j,k} \cdot I_{\text{sample}}(i, j). \quad (7)$$

This formulation ensures that in regions where  $w_{i,j,k}$  is high (e.g., textured areas), the tensor retains the original pixel details; where  $w_{i,j,k}$  is low (e.g., smooth areas), the tensor relies more on the global LUT mapping, thereby preserving color consistency. The entire construction process is fully differentiable, enabling end-to-end training. The resulting six-dimensional tensor  $T$  compactly encapsulates both global and local information, forming the unified representation for the subsequent manifold-adaptive sampling step.

#### E. Manifold-Adaptive High-Dimensional Sampling

Given the constructed six-dimensional fusion tensor  $T$ , the final enhanced output for each pixel is obtained through a manifold-adaptive high-dimensional sampling mechanism that dynamically adjusts the interpolation kernel based on local edge orientation, texture strength, and color similarity. For a pixel with spatial coordinates  $(x, y)$ , linear RGB values  $(r, g, b)$ , and luminance  $l$  (computed as  $l = 0.299r + 0.587g +$

$0.114b$ ), we first compute its continuous indices in each dimension:

$$x_g = \frac{x}{W} \cdot (N_x - 1), \quad (8a)$$

$$y_g = \frac{y}{H} \cdot (N_y - 1), \quad (8b)$$

$$l_g = l \cdot (N_l - 1), \quad (8c)$$

$$r_g = r \cdot (N_c - 1), \quad (8d)$$

$$g_g = g \cdot (N_c - 1), \quad (8e)$$

$$b_g = b \cdot (N_c - 1), \quad (8f)$$

where  $W$  and  $H$  are the image width and height, respectively. These real-valued indices are then decomposed into integer and fractional parts as in standard interpolation.

Unlike conventional trilinear interpolation, which always uses the eight surrounding vertices in the spatial-luminance dimensions, our manifold-adaptive scheme introduces three key improvements: (1) local tangent space estimation, (2) adaptive neighbor selection based on edge orientation, and (3) anisotropic weighting that respects the image manifold.

1) *Local Tangent Space Estimation*: From the luminance map  $L$ , we compute the gradient angle  $\theta = \arctan(\partial L / \partial y, \partial L / \partial x)$  and edge strength  $e = \|\nabla L\|$ . The edge-aware uncertainty  $u = \exp(-\beta e)$  from Eq. 3 is reused. The anisotropic scaling factor along the edge direction is defined as  $\gamma = 1 + \kappa \cdot u$ , where  $\kappa$  controls the elongation (set to 3.0 in practice). This factor allows the sampling kernel to stretch along the edge while remaining compact across it.

2) *Adaptive Neighbor Selection*: Instead of always using all eight spatial-luminance corners  $(x_{0/1}, y_{0/1}, l_{0/1})$ , we compute a confidence mask for each corner based on whether the line segment from the pixel to that corner crosses a strong edge. For a corner with spatial offset  $(\Delta x, \Delta y)$ , we evaluate the maximum gradient magnitude along the line segment. If the maximum exceeds a threshold  $\tau$  (set to 0.15), the corner is excluded. Typically, for a pixel on an edge, only 4–6 corners are retained, all lying on the same side of the edge. This prevents interpolation from mixing pixels from different objects, eliminating ghosting artifacts.

3) *Manifold Weight Computation*: For each selected corner  $c$  with spatial-luminance coordinates  $(x_c, y_c, l_c)$  and corresponding tensor value  $T_c$ , we compute the sampling weight as:

$$w_c = \exp\left(-\frac{\|\Delta \mathbf{p}_c\|^2}{2\sigma_s^2}\right) \cdot \exp\left(-\frac{\|\Delta \mathbf{rgb}_c\|^2}{2\sigma_r^2}\right) \cdot (1 - u_c), \quad (9)$$

where  $\Delta \mathbf{p}_c = ((x_c - x_g)/\gamma_x, (y_c - y_g)/\gamma_y, l_c - l_g)$  is the spatial-luminance distance anisotropically scaled by  $\gamma$  along the edge direction,  $\Delta \mathbf{rgb}_c = (r_c - r_g, g_c - g_g, b_c - b_g)$  is the RGB difference,  $\sigma_s = 0.5$  and  $\sigma_r = 0.2$  are bandwidth parameters, and  $u_c$  is the edge uncertainty at the corner. The weights are normalized:  $\mathbf{w}_c = \mathbf{w}_c / \sum_c w_c$ .

4) *Color Subspace Interpolation and Final Aggregation*: For each selected corner  $c$ , we first perform trilinear interpolation over the RGB dimensions using the eight surrounding

total regularization loss combined these two terms:

$$L_{TV} = \lambda_s R_{\text{spatial}} + \lambda_l R_{\text{luma}}, \quad (6)$$

其中  $\lambda_s$  和  $\lambda_l$  是标量权重（实践中设为0.01）。通过将  $L_{TV}$  纳入训练目标，网络学习生成在平滑区域空间连贯、同时又能自适应图像结构的权重网格，从而有效抑制伪影并保持清晰细节。

#### D. 六维融合张量的构建

得到双边权重网格  $W$  和可学习三维查找表  $LUT \in \mathbb{R}^{N_c \times N_c \times N_c \times 3}$  后，我们现在构建一个六维融合张量  $T$ ，它联合编码全局色彩映射和空间变化细节保持。张量  $T$  定义在六个维度上：空间坐标  $X$  和  $Y$ 、亮度级  $L$  以及 RGB 颜色索引  $R, G, B$ 。其形状为  $\mathbb{R}^{N_x \times N_y \times N_l \times N_c \times N_c \times N_c \times 3}$ ，其中  $N_x, N_y$  表示空间网格分辨率， $N_l$  表示亮度分箱数量， $N_c$  表示三维查找表尺寸。最后一个维度存储每种索引组合对应的输出线性 RGB 值。

为了平衡计算效率与复原质量，我们为高分辨率遥感图像设置了  $N_x = 128$ 、 $N_y = 72$ 、 $N_l = 24$  和  $N_c = 33$ 。空间下采样因子（约  $30\times$ ）将张量大小减少到约 193.5 万个参数，而亮度维度和查找表尺寸为精确的色彩风格化和细节自适应提供了足够的粒度。

张量通过在每个网格节点混合两种信息源来填充：来自三维查找表的全局色彩映射和来自输入图像的原始像素值。对于空间索引为  $(i, j)$  且亮度索引为  $k$  的给定网格节点，我们首先在对应网格单元的中心对线性 RGB 图像  $I_{lin}$  进行采样，得到  $I_{\text{sample}}(i, j) \in \mathbb{R}^3$ 。来自  $W$  的双边权重  $w_{i,j,k} \in [0, 1]$  控制局部细节的贡献。颜色索引  $(r, g, b)$  对应的张量项定义如下：

$$T(i, j, k, r, g, b) = (1 - w_{i,j,k}) \cdot LUT(r, g, b) + w_{i,j,k} \cdot I_{\text{sample}}(i, j). \quad (7)$$

该公式确保在  $w_{i,j,k}$  较高的区域（如纹理区域），张量保留原始像素细节；在  $w_{i,j,k}$  较低的区域（如平滑区域），张量更多依赖全局 LUT 映射，从而保持颜色一致性。整个构建过程完全可微，支持端到端训练。生成的六维张量  $T$  紧凑地封装了全局和局部信息，构成后续流形自适应采样步骤的统一表示。

#### E. 流形自适应高维采样

给定构建的六维融合张量  $T$ ，每个像素的最终增强输出通过流形自适应高维采样机制获得，该机制根据局部边缘方向、纹理强度和颜色相似性动态调整插值核。对于一个具有空间坐标  $(x, y)$ 、线性 RGB 值  $(r, g, b)$  和亮度  $l$ （计算为  $l = 0.299r + 0.587g +$

$0.114b$ ），我们首先计算其在每个维度上的连续索引：

$$x_g = \frac{x}{W} \cdot (N_x - 1), \quad (8a)$$

$$y_g = \frac{y}{H} \cdot (N_y - 1), \quad (8b)$$

$$l_g = l \cdot (N_l - 1), \quad (8c)$$

$$r_g = r \cdot (N_c - 1), \quad (8d)$$

$$g_g = g \cdot (N_c - 1), \quad (8e)$$

$$b_g = b \cdot (N_c - 1), \quad (8f)$$

其中  $W$  和  $H$  分别为图像宽度和高度。这些实值索引随后被分解为整数部分和小数部分，与标准插值中的做法相同。

与始终使用空间-亮度维度中八个周围顶点的传统三线性插值不同，我们的流形自适应方案引入了三项关键改进：

(1) 局部切空间估计，(2) 基于边缘方向的自适应邻域选择，以及 (3) 尊重图像流形的各向异性加权。

1) 局部切空间估计：从亮度图  $L$  中，我们计算梯度角  $\theta = \arctan(\partial L / \partial y, \partial L / \partial x)$  和边缘强度  $e = \|\nabla L\|$ 。来自式 3 的边缘感知不确定性  $u = \exp(-\beta e)$  被复用。沿边缘方向各向异性缩放因子定义为  $\gamma$ ，其中控制伸长程度（实际设为 3.0）。该因子使采样核沿边缘方向拉伸，同时在垂直于边缘的方向上保持紧凑。

2) 自适应邻域选择：并非总是使用全部八个空间-亮度角点  $(x_{0/1}, y_{0/1}, l_{0/1})$ ，而是为每个角点计算一个置信度掩码，依据是从像素到该角点的线段是否穿过强边缘。对于空间偏移为  $(\Delta x, \Delta y)$  的角点，我们评估沿该线段的最大梯度幅值。若最大值超过阈值  $\tau$ （设为 0.15），则排除该角点。通常，对于边缘上的像素，仅保留 4–6 个角点，且它们全部位于边缘的同一侧。这防止了插值混合来自不同物体的像素，消除了鬼影伪影。

3) 流形权重计算：对于每个选定的角点  $c$ ，其空间-亮度坐标为  $(x_c, y_c, l_c)$ ，对应的张量值为  $T_c$ ，我们按如下方式计算采样权重：

$$w_c = \exp\left(-\frac{\|\Delta \mathbf{p}_c\|^2}{2\sigma_s^2}\right) \cdot \exp\left(-\frac{\|\Delta \mathbf{rgb}_c\|^2}{2\sigma_r^2}\right) \cdot (1 - u_c), \quad (9)$$

其中  $\Delta \mathbf{p}_c = ((x_c - x_g)/\gamma_x, (y_c - y_g)/\gamma_y, l_c - l_g)$  是沿边缘方向由  $\gamma$  各向异性缩放后的空间-亮度距离， $\Delta \mathbf{rgb}_c = (r_c - r_g, g_c - g_g, b_c - b_g)$  是 RGB 差异， $\sigma_s = 0.5$  和  $\sigma_r = 0.2$  是带宽参数， $u_c$  是该角点处的边缘不确定性。权重经过归一化： $w_c = w_c / \sum_c w_c$ 。

4) 颜色子空间插值与最终聚合：对于每个选定的角点  $c$ ，我们首先利用周围的八个

LUT vertices (same as Eq. 8) to obtain an intermediate color  $\mathbf{c}_c$ . Then the final pixel value is the weighted sum:

$$c_{xy} = \sum_c \tilde{w}_c \cdot c_c. \quad (10)$$

The resulting value  $\mathbf{c}_{xy}$  is the enhanced linear RGB value for the pixel at  $(x, y)$ . It is then converted back to the sRGB color space using the inverse of the linearization function, yielding the final dehazed pixel. This manifold-adaptive sampling is fully differentiable and adds only a small computational overhead (approximately 0.05 ms per 1080p image) because the neighbor selection and weight computation are implemented via efficient grid-based pre-filtering.

5) *Training Objectives*: To train 6<sup>th</sup>Grid-Net, we adopt a composite loss function that jointly optimizes pixel-wise accuracy, perceptual quality, color fidelity, and grid smoothness. The total loss is defined as:

$$L_{\text{total}} = \lambda_{l1} L_{l1} + \lambda_{\text{perc}} L_{\text{perc}} + \lambda_{\text{col}} L_{\text{col}} + \lambda_{\text{tv}} L_{\text{TV}}, \quad (11)$$

where  $\lambda_{l1}$ ,  $\lambda_{\text{perc}}$ ,  $\lambda_{\text{col}}$ , and  $\lambda_{\text{tv}}$  are balancing hyperparameters (set to 1.0, 0.1, 0.1, and 0.01, respectively).

L1 Loss  $L_{l1}$ : Computed in the linear RGB space between the predicted output  $\hat{\mathbf{I}}_{\text{lin}}$  and the ground truth  $\mathbf{I}_{\text{lin}}^{\text{gt}}$ , encouraging precise pixel-level reconstruction:

$$L_{l1} = \frac{1}{N} \sum \left\| \hat{\mathbf{I}}_{\text{lin}} - \mathbf{I}_{\text{lin}}^{\text{gt}} \right\|_1, \quad (12)$$

Perceptual Loss  $L_{\text{perc}}$ : Applied in the sRGB space using a pre-trained VGG-16 network to measure feature-level similarity, improving visual quality:

$$L_{\text{perc}} = \frac{1}{C} \sum_l \left\| \phi_l(\hat{\mathbf{I}}_{\text{srgb}}) - \phi_l(\mathbf{I}_{\text{srgb}}^{\text{gt}}) \right\|_1. \quad (13)$$

where  $\phi_l$  denotes the feature map from the  $l$ -th layer of VGG-16.

Color Loss  $L_{\text{col}}$ : Defined in the LAB color space to enhance color accuracy. The L1 distance is computed separately for the luminance ( $L$ ) and chrominance ( $a$ ,  $b$ ) channels:

$$L_{\text{col}} = \frac{1}{N} \sum \left\| \mathbf{L}_{\text{out}} - \mathbf{L}_{\text{gt}} \right\|_1 + \frac{1}{N} \sum \left\| \mathbf{a}_{\text{out}} - \mathbf{a}_{\text{gt}} \right\|_1 + \frac{1}{N} \sum \left\| \mathbf{b}_{\text{out}} - \mathbf{b}_{\text{gt}} \right\|_1. \quad (14)$$

TV Regularization Loss  $L_{\text{TV}}$ : Enforces smoothness on the bilateral weight grid  $\mathbf{W}$  while preserving edges, as detailed in Section III-C. It combines spatial and luminance total variation terms:

$$L_{\text{TV}} = \lambda_s R_{\text{spatial}} + \lambda_l R_{\text{luma}}, \quad (15)$$

where  $R_{\text{spatial}}$  incorporates edge-aware weighting to prevent over-smoothing at boundaries, and  $R_{\text{luma}}$  ensures coherent variation across luminance bins.

The composite loss is minimized end-to-end using the Adam optimizer, enabling the network to simultaneously learn the 3D LUT, the weight generation UNet, and the fusion tensor parameters, thereby achieving high-quality dehazing with both global color consistency and sharp local details.

To more clearly illustrate the computational procedure of the proposed manifold-adaptive high-dimensional sampling, Algorithm ?? presents a PyTorch-style pseudo-code implementation of its core steps.

TABLE I  
PSEUDO-CODE OF MANIFOLD-ADAPTIVE HIGH-DIMENSIONAL SAMPLING (PYTORCH STYLE)

| Step | Operation                                                                                    | Description                                                                                                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Input: hazy image, img, lut, W, 3D LUT, bilateral (Nx,Ny,Nl,Nc) weight grid, grid parameters |                                                                                                                                                                                                                                                |
| 2    | Compute                                                                                      | $\mathbf{xg} = \mathbf{x}/\mathbf{W} * (\mathbf{Nx}-1)$ ; $\mathbf{yg} = \mathbf{y}/\mathbf{H} * (\mathbf{Ny}-1)$ ; $\mathbf{lg} = (\text{spatial, luminance, l} * (\mathbf{Nl}-1))$ ; $\mathbf{rg} = \mathbf{r} * (\mathbf{Nc}-1)$ ; RGB)     |
| 3    | Estimate                                                                                     | $\text{localtheta, e} = \text{grad}(\text{lum})$ ; $\text{u tangent space} = \exp(-\text{beta} * \mathbf{e})$ ; $\text{gamma} = (\text{edge orientation, } 1 + \text{kappa} * \text{u strength, uncertainty})$ and anisotropic factor          |
| 4    | Adaptive selection: keep corners on same side of edge                                        | $\text{neighbor mask} = []$                                                                                                                                                                                                                    |
|      | for corner in corners:                                                                       | if $\text{max\_grad\_along\_segment}(\text{corner}) < \text{tau}$ : $\text{mask.append}(\text{corner})$                                                                                                                                        |
| 5    | Compute manifold                                                                             | $\mathbf{w\_c} = \text{weight for each se} \cdot \exp(-\ \mathbf{dp}\ ^2 / (2 * \sigma_s^2))$<br>lected corner (spa- * $\exp(-\ \mathbf{drgb}\ ^2 / (2 * \sigma_r^2))$ )<br>tial distance + RGB * (1-u_c)<br>similarity + edge<br>uncertainty) |
| 6    | Trilinear interpolation                                                                      | $\text{interpola\_c\_c} = \text{tion in RGB sub-trilinear\_interp}(\text{lut, space per corner rg, gg, bg})$                                                                                                                                   |
| 7    | Final aggregation                                                                            | $\text{out} = \text{sum}(\mathbf{w\_c} * \mathbf{c\_c}) / \text{weighted sum sum}(\mathbf{w\_c})$                                                                                                                                              |

## IV. EXPERIMENTAL

### A. Experimental Settings

1) *Datasets*: 6<sup>th</sup>Grid-Net is evaluated on a unified dataset comprising two public benchmarks, SateHaze1K and RICE. In addition, the color recovery capability of 6<sup>th</sup>Grid-Net has been validated on the public dataset ViCoW.

SateHaze1K is a publicly available dataset for remote sensing image dehazing. The dataset integrates multi-sensor data and consists of 1,200 pairs of 640×640 images, each pair containing a hazy RGB image and the corresponding clear ground truth. The images are categorized into three levels of haze concentration: thin, moderate, and thick, with 400 pairs in each category. For each category, the images are split into 320 pairs for training, 40 pairs for validation, and 40 pairs for testing.

The Remote sensing Image Cloud rEmoving(RICE) dataset is a publicly available dataset for cloud removal in remote sensing images. The dataset consists of two subsets: RICE-I and RICE-II. The RICE-I subset is selected, which contains 500 image pairs of 512×512 pixels, each comprising a cloud-contaminated image and its corresponding cloud-free ground truth. Of these, 400 pairs are used for training, 50 pairs for validation, and 50 pairs for testing.

LUT 顶点（与公式8相同）在 RGB 维度上进行三线插值，以获得中间颜色 $\mathbf{c}_c$ 。然后，最终像素值为加权求和：

$$c_{xy} = \sum_c \tilde{w}_c \cdot c_c. \quad (10)$$

所得的值  $\mathbf{c}_{xy}$  即为像素  $(x,y)$  处增强后的线性RGB值。随后利用线性化函数的逆变换将其转换回sRGB色彩空间，生成最终的去雾像素。这种流形自适应采样是完全可微的，并且由于邻域选择与权重计算通过高效的基于网格的预滤波实现，仅增加了极小的计算开销（每张1080p图像约0.05毫秒）。

5) 训练目标：为训练 6<sup>th</sup>Grid-Net，我们采用一种复合损失函数，联合优化逐像素精度、感知质量、色彩保真度和网格平滑度。总损失定义如下：

$$L_{\text{total}} = \lambda_{l1} L_{l1} + \lambda_{\text{perc}} L_{\text{perc}} + \lambda_{\text{col}} L_{\text{col}} + \lambda_{\text{tv}} L_{\text{TV}}, \quad (11)$$

其中  $\lambda_{l1}$ 、 $\lambda_{\text{perc}}$ 、 $\lambda_{\text{col}}$  和  $\lambda_{\text{tv}}$  为平衡超参数（分别设为 1.0、0.1、0.1和0.01）。

L1 损失  $L_{l1}$ ：在线性 RGB 空间中计算预测输出  $\hat{\mathbf{I}}_{\text{lin}}$  与真实值  $\mathbf{I}_{\text{lin}}^{\text{gt}}$  之间的差异，鼓励精确的像素级重建：

$$L_{l1} = \frac{1}{N} \sum \left\| \hat{\mathbf{I}}_{\text{lin}} - \mathbf{I}_{\text{lin}}^{\text{gt}} \right\|_1, \quad (12)$$

感知损失  $L_{\text{perc}}$ ：在 sRGB 空间中使用预训练 VGG-16 网络来度量特征级相似度，从而提升视觉质量：

$$L_{\text{perc}} = \frac{1}{C} \sum_l \left\| \phi_l(\hat{\mathbf{I}}_{\text{srgb}}) - \phi_l(\mathbf{I}_{\text{srgb}}^{\text{gt}}) \right\|_1. \quad (13)$$

其中  $\phi_l$  表示来自 VGG-16 第  $l$  层的特征图。

颜色损失  $L_{\text{col}}$ ：在 LAB 色彩空间中定义，以增强颜色精度。L1 距离分别针对亮度（L）和色度（ $a, b$ ）通道进行计算：

$$L_{\text{col}} = \frac{1}{N} \sum \left\| \mathbf{L}_{\text{out}} - \mathbf{L}_{\text{gt}} \right\|_1 + \frac{1}{N} \sum \left\| \mathbf{a}_{\text{out}} - \mathbf{a}_{\text{gt}} \right\|_1 + \frac{1}{N} \sum \left\| \mathbf{b}_{\text{out}} - \mathbf{b}_{\text{gt}} \right\|_1. \quad (14)$$

全变分正则化损失  $L_{\text{TV}}$ ：对双边权重网格  $\mathbf{W}$  施加平滑性约束，同时保留边缘，详见第 III-C 节。它结合了空间全变分项和亮度全变分项：

$$L_{\text{TV}} = \lambda_s R_{\text{spatial}} + \lambda_l R_{\text{luma}}, \quad (15)$$

其中  $R_{\text{spatial}}$  结合了边缘感知加权以防止边界处的过度平滑， $R_{\text{luma}}$  确保亮度区间之间变化的一致连贯性。

使用Adam优化器对复合损失进行端到端最小化，使网络能够同时学习三维查找表、权重生成UNet和融合张量参数，从而实现具有全局颜色一致性和清晰局部细节的高质量去雾。

为了更清楚地说明所提出的流形自适应高维采样的计算过程，算法?? 给出了其核心步骤的PyTorch风格伪代码实现。

表I 流形自适应高维采样伪代码（PyTorch风格）

| 步骤 | 操作                                                        | 描述                                                                                                                                                                                                                             |
|----|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | 输入：有雾图像, img, lut, 3D LUT, 双边 (Nx, Ny, Nl, Nc) 权重网格, 网格参数 |                                                                                                                                                                                                                                |
| 2  | 计算                                                        | $\mathbf{xg} = \mathbf{x}/\mathbf{W} * (\mathbf{Nx}-1)$ ; $\mathbf{yg} = \mathbf{y}/\mathbf{H} * (\mathbf{Ny}-1)$ ; $\mathbf{lg} = (\text{空间, 亮度, l} * (\mathbf{Nl}-1))$ ; $\mathbf{rg} = \mathbf{r} * (\mathbf{Nc}-1)$ ; RGB) |
| 3  | 估计                                                        | $\text{局部theta, e} = \text{grad}(\text{lum})$ ; $\text{u 切空间} = \exp(-\text{beta} * \mathbf{e})$ ; $\text{gamma} = (\text{边缘 方向, } 1 + \text{kappa} * \text{u 强度, 不确定性})$ and 各向异性因子                                           |
| 4  | 自适应选择：保留角-点在同一侧                                           | $\text{邻域掩码} = []$                                                                                                                                                                                                             |
|    | 对于角点在                                                     | 角点中: if $\text{max\_grad\_along\_segment}(\text{角点}) < \text{tau}$ : $\text{mask.append}(\text{corner})$                                                                                                                       |
| 5  | 计算                                                        | 流形权重 = 每个 se 的权重为 $\exp(-\ \mathbf{dp}\ ^2 / (2 * \sigma_s^2))$<br>选中的 角点 (空- * $\exp(-\ \mathbf{drgb}\ ^2 / (2 * \sigma_r^2))$ )<br>空间距离 + RGB * (1-u_c)<br>相似性 + edge<br>不确定性)                                               |
| 6  | RGB 中的三线插值                                                | 子三线插值 (lut, 每角空间 rg, gg, bg)                                                                                                                                                                                                   |
| 7  | 最终聚合：输出                                                   | $= \text{和}(\mathbf{w\_c} * \mathbf{c\_c}) / \text{加权求和 sum}(\mathbf{w\_c})$                                                                                                                                                   |

## IV. EXPERIMENTAL

### A. 实验设置

1) 数据集：6<sup>th</sup>Grid-Net在一个由两个公开基准 SateHaze1K和RICE组成的统一数据集上进行了评估。此外，6<sup>th</sup>Grid-Net的颜色恢复能力已在公开数据集ViCoW上得到验证。

SateHaze1K是一个公开可用的遥感图像去雾数据集。该数据集整合了多传感器数据，包含1,200对 640×640 图像，每对包含一幅有雾RGB图像和对应的清晰真值。这些图像按雾浓度分为三个等级：薄、中等和厚，每个类别各400对。对于每个类别，图像被划分为320对用于训练，40对用于验证，40对用于测试。

遥感图像去云（RICE）数据集是一个公开可用的遥感图像云去除数据集。该数据集包含两个子集：RICE-I 和 RICE-II。本文选用 RICE-I 子集，其中包含 500 对大小为 512×512 像素的图像对，每对由一张被云污染的图像及其对应的无云真值组成。其中，400 对用于训练，50 对用于验证，50 对用于测试。

**A unified dataset combining SateHaze1K and RICE was constructed**, consisting of 1,700 image pairs in total: 500 cloudy pairs from RICE, along with 400 thick haze, 400 moderate haze, and 400 thin haze pairs from SateHaze1K. Each pair includes a remote sensing image to be processed and its corresponding clear ground-truth image. Among these, 1,360 pairs were used for training, 170 pairs for validation, and 170 pairs for testing, with all images uniformly resized to  $512 \times 512$ .

The ViCoW [9] dataset is a collection of image pairs specifically designed for historical image restoration and colorization tasks, with a focus on 20th-century Vietnam War-era film footage. It contains a total of 1,896 high-resolution ( $1280 \times 720$ ) image pairs, extracted from four historically significant Vietnamese films set during the Vietnam War era. The dataset is organized into training, validation, and test sets, enabling researchers to train and assess deep learning models for restoring and colorizing historical imagery. Each pair consists of an original color frame and its corresponding grayscale version, generated using the ITU-R BT.601 luminance formula.

2) *Implementation Detail*: 6<sup>th</sup>Grid-Net adopts an encoder-decoder architecture for bilateral weight grid generation. The U-Net comprises four encoding levels with channel dimensions 32, 64, 128, 256, each followed by a max-pooling layer. The decoder symmetrically upsamples the features and concatenates them with the corresponding encoder outputs via skip connections. The output of the final decoder layer is bilinearly interpolated to the target grid resolution of  $N_x = 128$ ,  $N_y = 72$ , and  $N_l = 24$ , followed by a  $3 \times 3$  convolution with Sigmoid activation to produce the grid weight  $\mathbf{W} \in [0, 1]^{128 \times 72 \times 24}$ . The 3D LUT is initialized as an identity mapping with size  $N_c = 33$ . The model is optimized using the Adam optimizer ( $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$ , weight decay = 0), with an initial learning rate of  $1 \times 10^{-4}$ . A cosine annealing scheduler is employed to gradually reduce the learning rate over 100 epochs, and the batch size is set to 16.

For the unified remote sensing dataset, the model is trained for 100 epochs. The total loss function is a weighted combination of four components:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{ll}} \mathcal{L}_{\text{ll}} + \lambda_{\text{perc}} \mathcal{L}_{\text{perceptual}} + \lambda_{\text{color}} \mathcal{L}_{\text{color}} + \lambda_{\text{TV}} \mathcal{L}_{\text{TV}}, \quad (16)$$

where  $\mathcal{L}_{\text{ll}}$  is the L1 loss computed in the linear RGB space,  $\mathcal{L}_{\text{perceptual}}$  is the perceptual loss based on VGG-16 features,  $\mathcal{L}_{\text{color}}$  is the color loss in the LAB space, and  $\mathcal{L}_{\text{TV}}$  is the edge-aware total variation regularization. The corresponding weights are set to  $\lambda_{\text{ll}} = 1.0$ ,  $\lambda_{\text{perc}} = 0.1$ ,  $\lambda_{\text{color}} = 0.1$ , and  $\lambda_{\text{TV}} = 0.01$ . The TV loss itself consists of spatial and luminance components, each with a coefficient of 1.0, and the edge decay parameter is set to  $\beta = 10.0$ .

Data preprocessing first converts the input sRGB images to the linear RGB space using the standard gamma-curve transformation, followed by luminance computation with the coefficients (0.299, 0.587, 0.114). The only data augmentation applied during training is random cropping to  $256 \times 256$  patches. During testing, images are kept at their original resolution. No learning rate warm-up or early stopping is used.

TABLE II  
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE UNIFIED REMOTE SENSING DATASET.

| Method                          | PSNR↑ | SSIM↑ | LPIPS↓ | Params↓ | FLOPs↓  | Size↓   | FPS↑   |
|---------------------------------|-------|-------|--------|---------|---------|---------|--------|
| Diffusion [35]                  | 21.29 | 0.81  | 0.10   | 109.74  | 1979.12 | 418.61  | 0.44   |
| RSazeDiff [36]                  | 17.67 | 0.83  | 0.17   | 0.63    | 53.02   | 2.41    | 13.31  |
| Restormer [37]                  | 15.25 | 0.24  | 0.78   | 119.12  | 340.80  | 1028.55 | 3.70   |
| AOD-Net [38]                    | 16.48 | 0.80  | 0.24   | 0.00    | 0.54    | 0.01    | 309.34 |
| HyperHazeOff [39]               | 25.14 | 0.92  | 0.08   | 2.47    | 38.09   | 29.07   | 8.86   |
| LUTwithGrid [6]                 | 20.67 | 0.81  | 0.21   | 0.46    | 0.67    | 1.78    | 44.50  |
| CLUT-Net [40]                   | 20.83 | 0.80  | 0.20   | 0.29    | 0.08    | 1.12    | 166.26 |
| 6 <sup>th</sup> Grid-Net (ours) | 22.52 | 0.84  | 0.16   | 1.94    | 8.09    | 7.84    | 70.83  |

All baseline methods are retrained under the same protocol to ensure a fair comparison. After training, the UNet weights are quantized to INT8 using dynamic post-training quantization, while the 3D LUT remains in FP32, resulting in a model size of under 2MB and enabling real-time inference on mobile devices.

3) *Evaluation Metrics*: To comprehensively evaluate the restoration performance of 6<sup>th</sup>Grid-Net, we adopt three full-reference metrics for quantitative analysis. Peak Signal-to-Noise Ratio (PSNR) [29] measures pixel-level reconstruction accuracy, with higher values indicating better fidelity to the ground truth. Structural Similarity Index (SSIM) [29] evaluates the preservation of brightness, contrast, and structural information, where higher values reflect superior structural consistency. Learned Perceptual Image Patch Similarity (LPIPS) [30] quantifies perceptual quality using deep feature embeddings; lower LPIPS values indicate results that are closer to human visual perception.

For efficiency evaluation, we report the following metrics. Number of parameters (Params) [31] reflects the model complexity, with fewer parameters implying a lighter architecture. FLOPs (floating point operations) [32] measures the computational cost during inference, where lower FLOPs enable faster processing. Model size [33] refers to the storage space occupied by the saved model file, which directly impacts deployment feasibility on resource-limited devices. Frames per second (FPS) [34] is measured on a mainstream mobile device to evaluate real-time capability, with higher FPS indicating better suitability for edge deployment.

### B. Experimental Results

To comprehensively evaluate the proposed 6<sup>th</sup>Grid-Net, we conduct two sets of experiments. The first set compares our method against state-of-the-art (SOTA) remote sensing image dehazing and all-in-one restoration methods on the unified dataset. The second set validates the generalization capability of 6<sup>th</sup>Grid-Net in a challenging color recovery task: restoring RGB colors from grayscale natural scene images using the ViCoW dataset. All baseline methods are retrained under the same protocol for fair comparison.

1) *Comparison with State-of-the-Art Methods*: We compare the proposed 6<sup>th</sup>Grid-Net against seven representative methods on the unified remote sensing dataset. Table II reports the quantitative results in terms of reconstruction fidelity, perceptual quality, and efficiency. Among all competitors, our method

**构建了一个结合 SateHaze1K 和 RICE 的统一数据集**, 共包含 1,700 对图像: 来自 RICE 的 500 对有云图像对, 以及来自 SateHaze1K 的 400 对浓雾、400 对中度雾霾和 400 对薄雾图像对。每对图像包含一张待处理的遥感图像及其对应的清晰真值图像。其中, 1,360 对用于训练, 170 对用于验证, 170 对用于测试, 所有图像均统一调整尺寸为  $512 \times 512$ 。

ViCoW [9] 数据集是专为历史图像复原与着色任务而设计的图像对集合, 重点关注20世纪越南战争时期的电影影像。它包含总共1,896对高分辨率 ( $1280 \times 720$ ) 图像对, 取自四部具有历史意义的、以越南战争时期为背景的越南电影。该数据集划分为训练集、验证集和测试集, 使研究人员能够训练和评估用于历史影像复原与着色的深度学习模型。每对图像由原始彩色帧及其对应的灰度版本组成, 灰度版本使用 ITU-R BT.601亮度公式生成。

2) 实现细节: 6<sup>th</sup>Grid-Net采用编码器-解码器架构来生成双边权重网格。U-Net包含四个编码层级, 通道维度分别为32、64、128、256, 每个层级后接一个最大池化层。解码器对称地对特征进行上采样, 并通过跳跃连接将其与对应的编码器输出拼接。最终解码器层的输出被双线性插值到目标网格分辨率  $N_x = 128$ 、 $N_y = 72$ 和 $N_l = 24$ , 随后经过一个带sigmoid激活的  $3 \times 3$  卷积, 生成网格权重  $\mathbf{W} \in [0, 1]^{128 \times 72 \times 24}$ 。三维查找表初始化为大小为 $N_c = 33$ 的恒等映射。模型使用Adam优化器 ( $\beta_1 = 0.9$ 、 $\beta_2 = 0.999$ , 权重衰减 = 0) 进行优化, 初始学习率为  $1 \times 10^{-4}$ 。采用余弦退火调度器在100个轮次内逐步降低学习率, 批大小设置为16。

对于统一遥感数据集, 模型训练 100 个轮次。总损失函数是四个分量的加权组合:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{ll}} \mathcal{L}_{\text{ll}} + \lambda_{\text{perc}} \mathcal{L}_{\text{perceptual}} + \lambda_{\text{color}} \mathcal{L}_{\text{color}} + \lambda_{\text{TV}} \mathcal{L}_{\text{TV}}, \quad (16)$$

其中  $\mathcal{L}_{\text{ll}}$  是在线性 RGB 空间中计算的 L1 损失,  $\mathcal{L}_{\text{perceptual}}$  是基于 VGG-16 特征的感知损失,  $\mathcal{L}_{\text{color}}$  是 LAB 空间中的颜色损失,  $\mathcal{L}_{\text{TV}}$  是边缘感知全变分正则化。对应的权重分别设置为  $\lambda_{\text{ll}} = 1.0$ 、 $\lambda_{\text{perc}} = 0.1$ 、 $\lambda_{\text{color}} = 0.1$ 和  $\lambda_{\text{TV}} = 0.01$ 。全变分损失本身由空间分量和亮度分量组成, 每个分量的系数均为 1.0, 边缘衰减参数设置为  $\beta = 10.0$ 。

数据预处理首先使用标准伽马曲线变换将输入的 s RGB 图像转换到线性 RGB 空间, 然后使用系数 (0.299,0.587,0.114) 进行亮度计算。训练期间应用的唯一数据增强是随机裁剪为  $256 \times 256$  图像块。在测试期间, 图像保持其原始分辨率。不使用学习率预热或早停。

表II 不同方法在统一遥感数据集上的定量比较。

| 方法                              | PSNR↑ | SSIM↑ | LPIPS↓ | 参数量↓   | 浮点运算次数↓ | 模型大小↓   | 每秒帧数↑  |
|---------------------------------|-------|-------|--------|--------|---------|---------|--------|
| 扩散模型 [35]                       | 21.29 | 0.81  | 0.10   | 109.74 | 1979.12 | 418.61  | 0.44   |
| RSazeDiff [36]                  | 17.67 | 0.83  | 0.17   | 0.63   | 53.02   | 2.41    | 13.31  |
| Restormer [37]                  | 15.25 | 0.24  | 0.78   | 119.12 | 340.80  | 1028.55 | 3.70   |
| AOD-Net [38]                    | 16.48 | 0.80  | 0.24   | 0.00   | 0.54    | 0.01    | 309.34 |
| HyperHazeOff [39]               | 25.14 | 0.92  | 0.08   | 2.47   | 38.09   | 29.07   | 8.86   |
| LUTwithGrid [6]                 | 20.67 | 0.81  | 0.21   | 0.46   | 0.67    | 1.78    | 44.50  |
| CLUT-Net [40]                   | 20.83 | 0.80  | 0.20   | 0.29   | 0.08    | 1.12    | 166.26 |
| 6 <sup>th</sup> Grid-Net (本文方法) | 22.52 | 0.84  | 0.16   | 1.94   | 8.09    | 7.84    | 70.83  |

所有基线方法都在相同的协议下重新训练, 以确保公平比较。训练后, UNet 权重使用动态训练后量化量化为 INT8, 而 3D LUT 保持为 FP32, 从而使模型大小低于 2MB, 并能在移动设备上实现实时推理。

3) 评估指标: 为了全面评估 6<sup>th</sup>Grid-Net的复原性能, 我们采用三种全参考指标进行定量分析。峰值信噪比 (PSNR) [29] 衡量像素级重建精度, 数值越高表示对真实值的保真度越好。结构相似性指数 (SSIM) [29] 评估亮度、对比度和结构信息的保持情况, 数值越高反映结构一致性越优。学习感知图像块相似度 (LPIPS) [30] 利用深度特征嵌入量化感知质量; LPIPS值越低, 表示结果越接近人类视觉感知。

在效率评估方面, 我们报告以下指标。参数量 (Params) [31] 反映模型复杂度, 参数量越少意味着架构越轻量。浮点运算次数 (FLOPs) [32] 衡量推理过程中的计算成本, FLOPs越低可加快处理速度。模型大小 [33] 指保存的模型文件所占的存储空间, 直接影响在资源受限设备上的部署可行性。每秒帧数 (FPS) [34] 在主流移动设备上测量, 用于评估实时能力, FPS越高表示越适合边缘部署。

### B. 实验结果

为全面评估所提出的 6<sup>th</sup>Grid-Net, 我们进行了两组实验。第一组实验在统一数据集上将我们的方法同最先进的遥感图像去雾与一体化复原方法进行比较。第二组实验通过使用 ViCoW 数据集从灰度自然场景图像中恢复 RGB 颜色这一具有挑战性的颜色恢复任务, 验证 6<sup>th</sup>Grid-Net 的泛化能力。所有基线方法均在相同协议下重新训练, 以保证公平比较。

1) 与最先进方法的比较: 我们在统一遥感数据集上, 将所提出的 6<sup>th</sup>Grid-Net与七种代表性方法进行了比较。表 II 报告了在重建保真度、感知质量和效率方面的定量结果。在所有比较方法中, 我们的方法

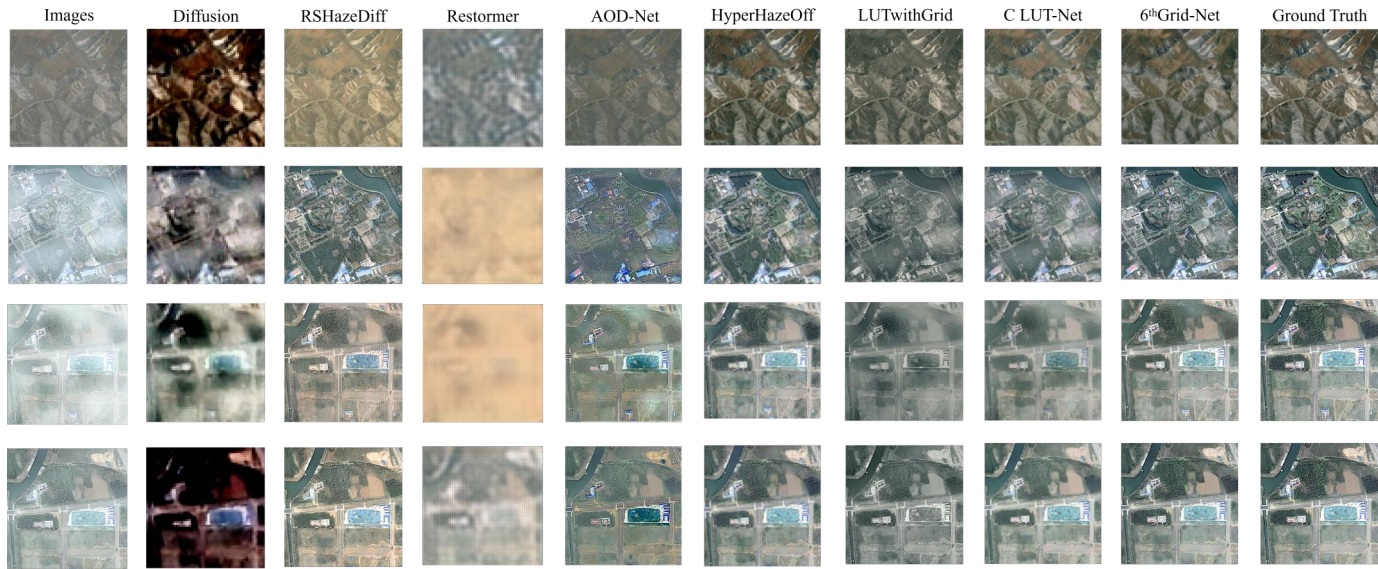


Fig. 3. **Qualitative results on remote sensing image dehazing.** Rows from top to bottom correspond to four degradation types: cloudy, moderate haze, thick haze, and thin haze, respectively. Columns from left to right show the input hazy image, the dehazed results produced by different baseline methods, the output of our proposed 6<sup>th</sup>Grid-Net (second last column), and the ground-truth clear image (last column). Compared with other methods, our 6<sup>th</sup>Grid-Net generates clearer textures, more faithful color restoration, fewer artifacts, and better edge preservation across all degradation levels.

TABLE III  
COLOR RECOVERY PERFORMANCE ON THE ViCoW DATASET.

| Method                   | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ |
|--------------------------|-----------------|-----------------|--------------------|
| AOD-Net                  | 28.10           | 0.96            | 0.14               |
| 6 <sup>th</sup> Grid-Net | <b>28.61</b>    | <b>0.97</b>     | <b>0.13</b>        |

achieves the highest PSNR (22.52 dB) and SSIM (0.836), as well as the lowest LPIPS (0.164), indicating superior pixel-wise accuracy, structural preservation, and perceptual naturalness. Although some lightweight models (e.g., AOD-Net, CLUT-Net) offer faster inference or smaller model size, they lag behind in restoration quality, especially under thick haze or cloud conditions. In contrast, 6<sup>th</sup>Grid-Net strikes a favorable trade-off between quality and efficiency, with 1.935M parameters and 8.09G FLOPs, while still running at 70.8 FPS on a mobile device.

Fig. 3 provides a qualitative comparison. It can be observed that methods such as Diffusion and RSHazeDiff produce oversmoothed outputs with residual haze, while Restormer and AOD-Net suffer from severe color distortion and blurring. LUTwithGrid and CLUT-Net recover reasonable colors but fail to preserve fine edges and textures. In comparison, 6<sup>th</sup>Grid-Net consistently reconstructs the sharpest details, most vivid colors, and cleanest backgrounds, effectively removing both thin and thick haze without introducing artifacts. These results validate the advantage of our manifold-adaptive high-dimensional sampling mechanism in jointly optimizing global color rendition and local edge preservation.

2) *Color Recovery from Grayscale Images*: To evaluate the generalization capability of our method beyond dehazing, we conduct color recovery experiments on the ViCoW dataset, which consists of grayscale-to-RGB image pairs extracted from historical Vietnam War-era film footage. Table III reports



Fig. 4. **Color recovery results on the ViCoW dataset.** 6<sup>th</sup>Grid-Net reconstructs vivid and semantically plausible colors, closely matching the ground truth while preserving fine structural details.

the quantitative results. Our method achieves a PSNR of 28.61 dB, an SSIM of 0.9653, and an LPIPS of 0.1326, indicating high-fidelity color reconstruction with strong structural and perceptual similarity to the ground truth.

Fig. 4 provides a qualitative comparison. For each of the three representative samples, the first column shows the input grayscale image, the second column presents the colorized output of our 6<sup>th</sup>Grid-Net, and the third column is the ground-truth color frame. It can be observed that our model not only recovers semantically plausible colors but also preserves fine textures and edges without introducing noticeable artifacts. The colorized results are visually close to the ground truth, demonstrating that the manifold-adaptive high-dimensional sampling mechanism, originally designed for dehazing, generalizes well to the task of grayscale image colorization. These results further confirm the versatility of our unified framework in handling diverse image restoration tasks.

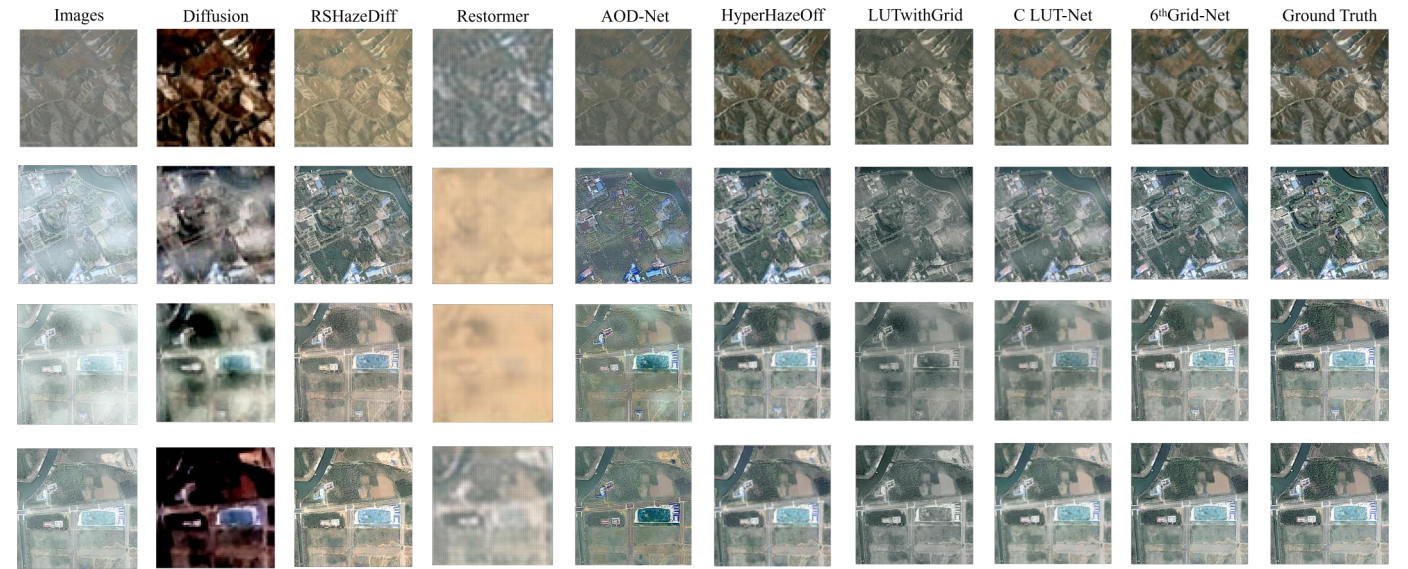


图3. 遥感图像去雾的定性结果。从上到下的行分别对应四种退化类型：多云、中度雾霾、浓雾和薄雾。从左到右的列分别显示输入的有雾图像、不同基线方法生成的去雾结果、我们提出的 6<sup>th</sup>Grid-Net的输出（倒数第二列）以及真实清晰图像（最后一列）。与其他方法相比，我们的 6<sup>th</sup>Grid-Net在所有退化级别上都能生成更清晰的纹理、更忠实的颜色恢复、更少的伪影和更好的边缘保持。

表III VICOW数据集上的颜色恢复性能。

| 方法                       | 峰值信噪比 $\uparrow$ | 结构相似性指数 $\uparrow$ | 学习感知图像块相似度 $\downarrow$ |
|--------------------------|------------------|--------------------|-------------------------|
| AOD-Net                  | 28.10            | 0.96               | 0.14                    |
| 6 <sup>th</sup> Grid-Net | <b>28.61</b>     | <b>0.97</b>        | <b>0.13</b>             |

取得了最高的峰值信噪比（22.52 dB）和结构相似性指数（0.836），以及最低的学习感知图像块相似度（0.164），表明其具有优越的逐像素精度、结构保持能力和感知自然度。尽管一些轻量化模型（如AOD-Net、CLUT-Net）提供了更快的推理速度或更小的模型大小，但它们在复原质量方面有所不足，尤其是在浓雾或云层条件下。相比之下，6<sup>th</sup>Grid-Net在质量和效率之间取得了良好的权衡，参数量为1.935M，浮点运算次数为8.09G，同时仍能在移动设备上以70.8每秒帧数运行。

图3 给出了定性比较。可以观察到，Diffusion 和 RSHazeDiff 等方法产生了过度平滑的输出，并伴有残余雾霾，而 Restormer 和 AOD-Net 则存在严重的颜色失真和模糊问题。LUTwithGrid 和 CLUT-Net 恢复了合理的颜色，但未能保留精细边缘和纹理。相比之下，6<sup>th</sup>Grid-Net 始终如一地重建出最清晰的细节、最鲜明的色彩和最干净的背景，能够有效去除薄雾和浓雾，且不引入伪影。这些结果验证了我们的流形自适应高维采样机制在联合优化全局色彩还原和局部边缘保持方面的优势。

2) 灰度图像的颜色恢复：为了评估我们方法在去雾之外的泛化能力，我们在 ViCoW 数据集上进行了颜色恢复实验，该数据集由从越战时期历史电影片段中提取的灰度到 RGB 图像对组成。表III报告了



图 4. ViCoW 数据集上的颜色恢复结果。6<sup>th</sup>Grid-Net 重建出鲜明且在语义上合理的颜色，与真实值高度吻合，同时保留了精细的结构细节。

定量结果。我们的方法实现了 28.61 dB 的 PSNR、0.9653 的 SSIM 和 0.1326 的 LPIPS，表明高保真颜色重建与真实值之间具有很强的结构相似性和感知相似性。

图4 给出了定性比较。对于三个代表性样本，第一列显示输入的灰度图像，第二列展示我们 6<sup>th</sup>Grid-Net的着色输出，第三列为真实彩色帧。可以观察到，我们的模型不仅恢复了语义上合理的颜色，还保留了精细纹理和边缘，且未引入可见伪影。着色结果在视觉上接近真实值，表明最初为去雾设计的流形自适应高维采样机制能够很好地推广到灰度图像着色任务。这些结果进一步证实了我们的统一框架在处理多样图像复原任务中的通用性。

TABLE IV  
QUANTITATIVE COMPARISON BEFORE AND AFTER DYNAMIC  
QUANTIZATION.

| Model              | PSNR↑ | SSIM↑ | LPIPS↓ | Params↓ | FLOPs↓      | Size↓       | FPS↑         |
|--------------------|-------|-------|--------|---------|-------------|-------------|--------------|
| Pre-quant. (FP32)  | 22.52 | 0.84  | 0.16   | 1.94    | 42.01       | 23.48       | 70.56        |
| Post-quant. (INT8) | 22.52 | 0.84  | 0.16   | 1.94    | <b>8.10</b> | <b>7.84</b> | <b>70.83</b> |

TABLE V  
ABLATION STUDY OF KEY COMPONENTS. PARAMS, FLOPS, AND MODEL  
SIZE REMAIN CONSTANT (1.935M, 8.092G, 7.84MB) ACROSS ALL  
VARIANTS.

| Variant | M-S | E-TV | BW | PSNR↑        | SSIM↑       | LPIPS↓      | FPS↑         |
|---------|-----|------|----|--------------|-------------|-------------|--------------|
| a       | ×   | ✓    | ✓  | 21.98        | 0.82        | 0.19        | 61.61        |
| b       | ✓   | ×    | ✓  | 21.95        | 0.812       | 0.19        | 60.82        |
| c       | ×   | ×    | ✓  | 20.67        | 0.82        | 0.17        | 59.13        |
| d       | ✓   | ✓    | ×  | 18.35        | 0.77        | 0.20        | 61.27        |
| Ours    | ✓   | ✓    | ✓  | <b>22.52</b> | <b>0.84</b> | <b>0.16</b> | <b>70.83</b> |

### C. Ablation Studies

1) *Quantitative ablation experiment*: To further validate the deployability of our method on resource-constrained edge devices, we apply dynamic post-training quantization to the UNet weights (INT8) while keeping the 3D LUT in FP32. Table IV compares the model before and after quantization. Notably, the quantized model retains exactly the same reconstruction quality (PSNR: 22.52 dB, SSIM: 0.8360, LPIPS: 0.1644) as its full-precision counterpart, while reducing the FLOPs from 42.007 G to 8.092 G (approximately 5.2× reduction) and the model size from 23.48 MB to 7.84 MB (approximately 3.0× reduction). Moreover, the inference speed slightly improves from 70.56 FPS to 70.83 FPS on a mobile device. These results demonstrate that our quantization strategy effectively compresses the model without sacrificing accuracy, making 6<sup>th</sup> Grid-Net well-suited for real-time remote sensing image dehazing on edge platforms.

2) *Component ablation experiment*: To validate each component, we compare five variants on the unified dataset (Table V). Variant (a) disables manifold-adaptive sampling (M-S); (b) removes edge-aware TV regularization (E-TV); (c) disables M-S, E-TV and replaces the bilateral weight grid (BW) with a fixed uniform weight; (d) removes BW; and Ours includes all three components. All variants share identical parameters, FLOPs and model size. As shown in Table V, removing M-S (variant a) reduces PSNR from 22.52 dB to 21.98 dB and increases LPIPS to 0.1857, confirming that M-S avoids cross-edge blur. Variant (b) (without E-TV) yields lower PSNR (21.95 dB) and higher LPIPS (0.1869), indicating that E-TV suppresses weight noise. Variant (c) performs significantly worse (PSNR 20.67 dB, LPIPS 0.1746) due to the absence of all adaptive mechanisms. Most critically, variant (d) (without BW) suffers the largest drop (PSNR 18.35 dB, SSIM 0.7739, LPIPS 0.2044), proving that the bilateral weight grid is indispensable for spatially varying fusion. The full model achieves the best performance across all metrics (PSNR 22.52 dB, SSIM 0.8360, LPIPS 0.1644) and the highest FPS (70.83). Thus, all three components contribute positively, with BW

being the most critical.

## V. DISCUSSION

### A. Visualization and Analysis of Bilateral Weight Grid

The core idea of the bilateral weight grid is to enable spatially-adaptive fusion between global LUT-based color mapping and original pixel details, thereby overcoming the mutual interference problem inherent in sequential pipelines (Fig. 6). Instead of applying a fixed blending factor across the entire image, our method learns a weight grid  $\mathbf{W} \in [0, 1]^{N_x \times N_y \times N_l}$  that controls, for each spatial location and luminance level, how much of the original detail should be preserved. As shown in Fig. 6, the weight heatmap (middle) and its overlay on the input image (right) clearly demonstrate that high weights (red) concentrate along edges and textured regions (e.g., building boundaries and vegetation), while low weights (blue) dominate in flat, homogeneous areas such as sky and haze. This behaviour directly addresses the limitation of sequential methods: in a “detail-then-color” pipeline, edge enhancement would distort the original color distribution, forcing the subsequent color rendition module to adapt to already-modified pixel values; conversely, a “color-then-detail” pipeline would have its edge-preserving capability weakened by the smoothing effect of color mapping. By learning a location- and luminance-dependent weight, our unified six-dimensional tensor decouples these two objectives: flat regions rely almost entirely on the LUT for consistent color styling, whereas detailed regions retain their original high-frequency information without being over-smoothed or artificially recolored.

The adaptability of the weight grid across luminance levels is further validated in Fig. 7. For a pixel located on a strong edge (e.g., building-sky boundary), the weight remains consistently high across all 24 luminance bins, ensuring that edge sharpness is preserved regardless of local brightness. In contrast, a pixel in a flat region (e.g., clear sky) maintains low weights throughout, indicating that the LUT alone provides sufficient color fidelity. A textured region (e.g., ground vegetation) exhibits moderate weights with only slight variation, reflecting that some local detail is beneficial but aggressive preservation is unnecessary. This consistent behaviour across luminance bins confirms that the model does not rely on a single brightness level to make blending decisions; instead, it learns a smooth, content-aware mapping along the luminance dimension, which avoids abrupt transitions that could otherwise introduce banding or halos.

Finally, Fig. 8 examines the spatial smoothness of the learned weight grid. The gradient magnitude map (right) reveals that the weight grid varies very little in homogeneous areas—e.g., the sky region appears nearly black—indicating that the edge-aware total variation regularization successfully suppresses noisy or blocky artifacts that would otherwise arise from unconstrained learning. Strong gradients only appear where the image itself contains salient edges (e.g., building contours), allowing the weight grid to change sharply exactly where detail preservation is needed. Without this regularization (ablated in Table V), the weight grid would exhibit scattered

表 IV 动态量化前后的定量比较。

| 模型         | 峰值信噪比↑ | 结构相似性指数↑ | 学习感知图像块相似度↓ | 参数量↓ | 浮点运算次数↓     | 模型大小↓       | 每秒帧数↑        |
|------------|--------|----------|-------------|------|-------------|-------------|--------------|
| 量化前 (FP32) | 22.52  | 0.84     | 0.16        | 1.94 | 42.01       | 23.48       | 70.56        |
| 量化后 (INT8) | 22.52  | 0.84     | 0.16        | 1.94 | <b>8.10</b> | <b>7.84</b> | <b>70.83</b> |

表 V 关键组件的消融实验。所有变体的参数量、浮点运算次数和模型大小保持不变（1.935M、8.092G、7.84MB）。

| 变体   | M-S | E-TV | BW | 峰值信噪比↑       | 结构相似性指数↑    | LPIPS↓      | FPS↑         |
|------|-----|------|----|--------------|-------------|-------------|--------------|
| a    | ×   | ✓    | ✓  | 21.98        | 0.82        | 0.19        | 61.61        |
| b    | ✓   | ×    | ✓  | 21.95        | 0.812       | 0.19        | 60.82        |
| c    | ×   | ×    | ✓  | 20.67        | 0.82        | 0.17        | 59.13        |
| d    | ✓   | ✓    | ×  | 18.35        | 0.77        | 0.20        | 61.27        |
| Ours | ✓   | ✓    | ✓  | <b>22.52</b> | <b>0.84</b> | <b>0.16</b> | <b>70.83</b> |

### C. 消融研究

1) 定量消融实验：为进一步验证我们的方法在资源受限的边缘设备上的可部署性，我们对UNet权重应用动态训练后量化（INT8），同时将3D LUT保持为FP32。表IV比较了量化前后的模型。值得注意的是，量化模型保持了与全精度模型完全相同的重建质量（PSNR: 22.52 dB, SSIM: 0.8360, LPIPS: 0.1644），同时将FLOPs从42.007 G降低至8.092 G（约5.2× 倍降低），模型大小从23.48 MB降至7.84 MB（约3.0×倍降低）。此外，在移动设备上推理速度从70.56 FPS略微提升至70.83 FPS。这些结果表明，我们的量化策略在无损精度的前提下有效压缩了模型，使6<sup>th</sup> Grid-Net非常适合在边缘平台上进行实时遥感图像去雾。

2) 组件消融实验：为了验证每个组件，我们在统一数据集上比较了五种变体（表V）。变体（a）禁用了流形自适应采样（M-S）；（b）移除了边缘感知总变差正则化（E-TV）；（c）禁用了M-S、E-TV，并将双边权重网格（BW）替换为固定均匀权重；（d）移除了BW；而Ours包含全部三个组件。所有变体的参数量、浮点运算次数和模型大小均相同。如表V所示，移除M-S（变体a）使PSNR从22.52 dB降至21.98 dB，LPIPS增至0.1857，证实了M-S可以避免跨边缘模糊。变体（b）（不含E-TV）的PSNR更低（21.95 dB），LPIPS更高（0.1869），表明E-TV可以抑制权重噪声。变体（c）由于缺少所有自适应机制，性能显著下降（PSNR 20.67 dB, LPIPS 0.1746）。最关键的是，变体（d）（不含BW）的下降幅度最大（PSNR 18.35 dB, SSIM 0.7739, LPIPS 0.2044），证明双边权重网格对于空间变化融合不可或缺。完整模型在所有指标上都取得了最佳性能（PSNR 22.52 dB, SSIM 0.8360, LPIPS 0.1644）和最高FPS（70.83）。因此，所有三个组件都做出了积极贡献，其中BW

最为关键。

## V. DISCUSSION

### A. 双边权重网络的可视化与分析

双边权重网络的核心思想是能够在基于全局LUT的色彩映射与原始像素细节之间实现空间自适应融合，从而克服顺序流水线中固有的相互干扰问题（图6）。我们没有在整个图像上应用固定的混合因子，而是学习了一个权重网格 $\mathbf{W} \in [0, 1]^{N_x \times N_y \times N_l}$ ，该网格针对每个空间位置和亮度级控制应保留多少原始细节。如图6所示，权重热图（中）及其在输入图像上的叠加（右）清楚地表明，高权重（红色）集中在边缘和纹理区域（例如建筑边界和植被），而低权重（蓝色）在天空和雾霾等平坦、均匀区域中占主导地位。这种行为直接解决了顺序方法的局限性：在“先细节后色彩”的流水线中，边缘增强会扭曲原始色彩分布，迫使后续的色彩再现模块适配已经修改的像素值；相反，“先色彩后细节”的流水线则会因色彩映射的平滑效果而削弱其保边能力。通过学习一个依赖于位置和亮度的权重，我们的统一六维张量解耦了这两个目标：平坦区域几乎完全依赖LUT来实现一致的色彩风格化，而细节区域则保留其原始高频信息，不会过度平滑或被不自然地重新着色。

权重网络在不同亮度级别上的适应性在图7中得到了进一步验证。对于位于强边缘（如建筑-天空边界）上的像素，其权重在所有24个亮度分箱中始终保持在较高水平，确保无论局部亮度如何，边缘锐度都能得到保留。相比之下，位于平坦区域（如晴空）的像素始终保持低权重，这表明仅靠LUT就足以提供足够的色彩保真度。纹理区域（如地面植被）则表现出中等权重，且仅有轻微变化，说明保留一定的局部细节是有益的，但无需进行过度保留。这种跨亮度分箱的一致行为证实了该模型并非依赖单一亮度级别来做出融合决策；相反，它沿着亮度维度学习了一种平滑的、内容感知的映射，从而避免了可能引入色带伪影或光晕的突变过渡。

最后，图8考察了所学习权重网络的空间平滑度。右侧的梯度幅值图显示，权重网络在均匀区域中变化极小——例如天空区域几乎呈黑色——这表明边缘感知全变分正则化成功抑制了不受约束学习可能产生的噪声或块状伪影。强梯度仅出现在图像本身包含显著边缘（如建筑物轮廓）的位置，使权重网络能够恰好在需要细节保持的地方发生急剧变化。若没有这种正则化（在表V中消融），权重网络将出现散布的

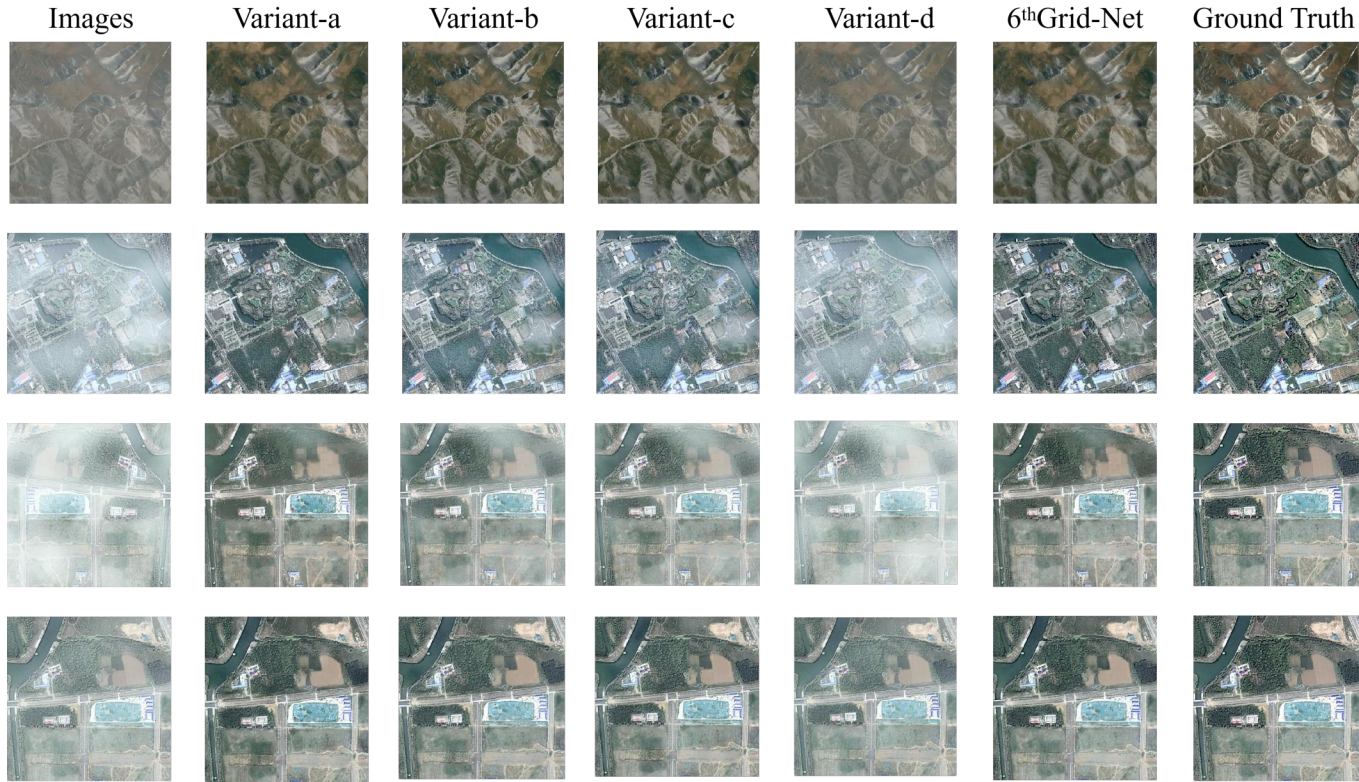


Fig. 5. **Ablation Results of Component Effectiveness.** Rows from top to bottom correspond to four degradation types: cloudy, moderate haze, thick haze, and thin haze, respectively. Columns from left to right show the input image, the processed results produced by different ablation variants (variants a–d as defined in Table V), the output of our complete model (second last column), and the ground-truth clear image (last column). Compared with the ablated variants, our full model achieves superior restoration with clearer textures, more faithful color restoration, fewer artifacts, and better edge preservation, confirming the necessity of each proposed component.

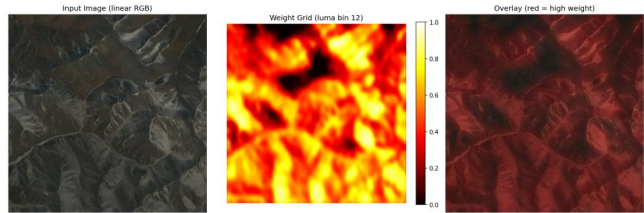


Fig. 6. **Visualization of the bilateral weight grid for a single luminance bin.** Left: input linear RGB image. Middle: weight heatmap (hot colormap) for the central luminance bin (index 12 out of 24), where red (value close to 1) indicates preservation of original details and blue (value close to 0) indicates reliance on global LUT mapping. Right: overlay of the weight heatmap on the input image, showing that high weights concentrate along edges and textured regions, while low weights dominate in flat areas.

high-frequency fluctuations, which could lead to unnatural textures or flickering in video applications. The proposed edge-guided smoothing constraint thus strikes a balance: it enforces spatial coherence in flat regions while preserving the ability to adapt to image structures, yielding a weight grid that is both interpretable and effective for high-quality dehazing.

#### B. Analysis of Manifold-Adaptive Sampling

Conventional high-dimensional interpolation methods, such as trilinear interpolation, apply a fixed isotropic kernel to every pixel regardless of local image structure. While computationally efficient, this strategy inevitably mixes information from both sides of a strong edge, leading to blurred boundaries,

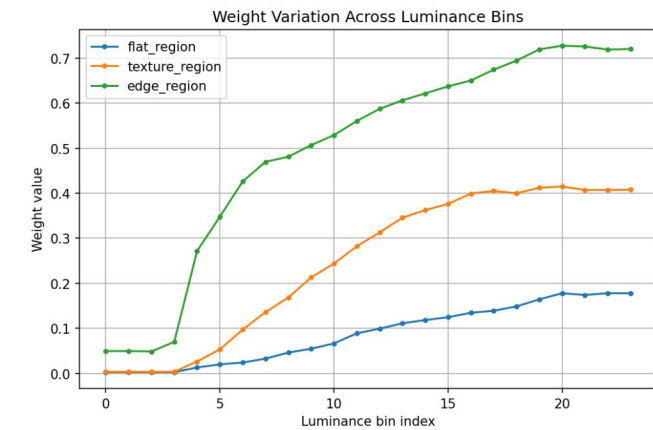


Fig. 7. **Weight values across luminance bins for three representative regions.** The curves correspond to a flat region, a textured region, and an edge region. The edge region maintains consistently high weights across all luminance levels to preserve structural details, whereas the flat region exhibits low weights throughout, indicating that the LUT-based color mapping is sufficient. The textured region shows moderate weights with a slight variation, demonstrating that the learned weight grid adapts to local image content.

halos, and color bleeding—artifacts that are particularly detrimental in remote sensing imagery where fine structural details (e.g., building outlines, road networks) must be preserved. To overcome this limitation, our method replaces the fixed kernel with a manifold-adaptive sampling mechanism that respects the intrinsic low-dimensional structure of natural

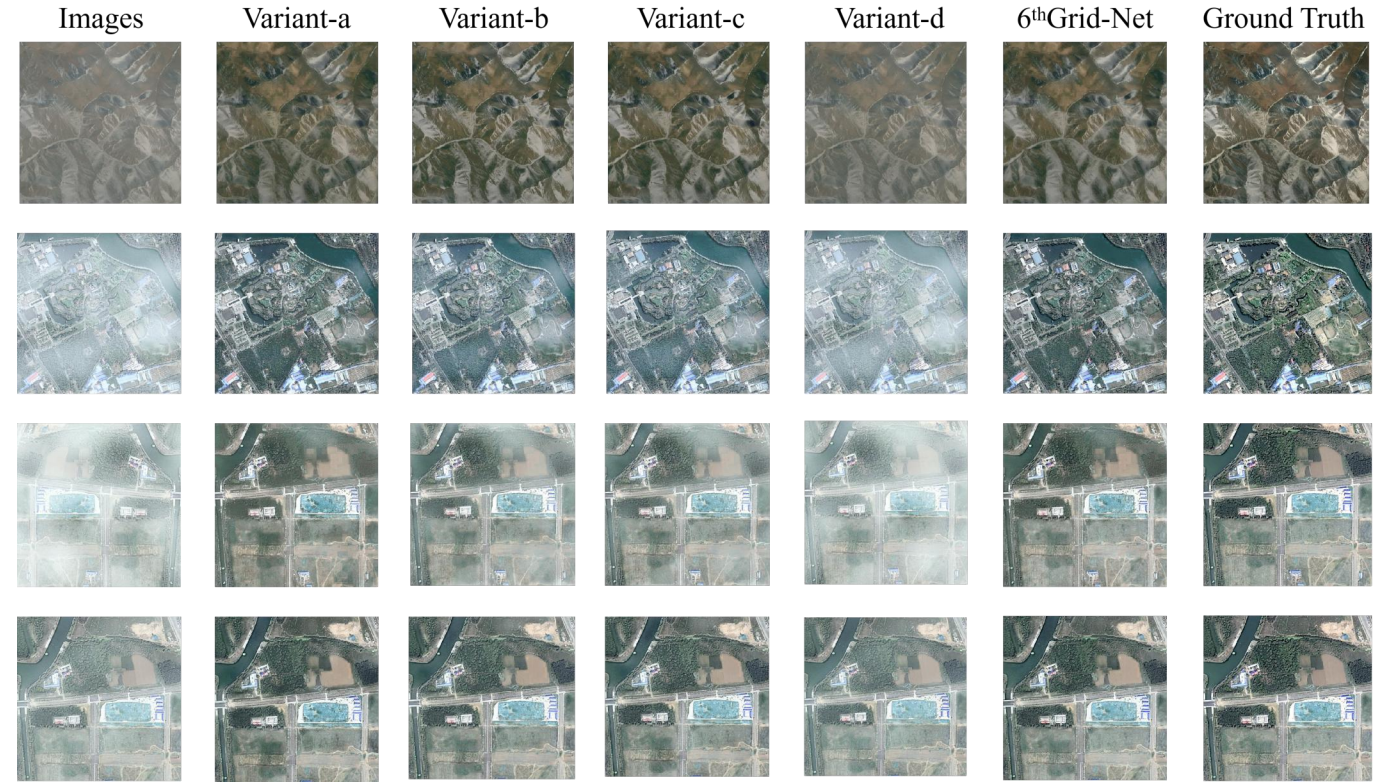


图5. 组件有效性的消融结果。自上而下各行分别对应四种退化类型：多云、中度雾霾、浓雾和薄雾。从左至右各列依次为输入图像、不同消融变体（表V中定义的变体a–d）生成的处理结果、我们完整模型的输出（倒数第二列）以及真实清晰图像（最后一列）。与各消融变体相比，我们的完整模型实现了更优的复原效果，具有更清晰的纹理、更忠实的颜色恢复、更少的伪影和更好的边缘保持，证实了每个所提组件的必要性。

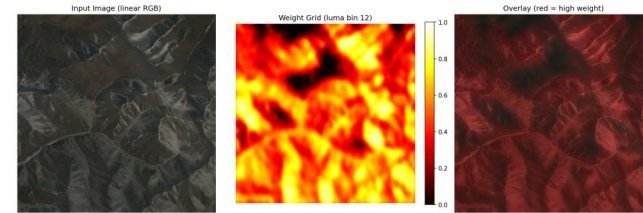


图6. 单个亮度分箱的双边权重网格可视化。左：输入的线性RGB图像。中：中央亮度分箱（24个分箱中的第12个）的权重热图（热力色图），其中红色（值接近1）表示保留原始细节，蓝色（值接近0）表示依赖全局LUT映射。右：叠加在输入图像上的权重热图，显示高权重集中在边缘和纹理区域，而低权重在平坦区域占主导。

高频波动，这可能导致视频应用中出现不自然的纹理或闪烁。因此，所提出的边缘引导平滑约束取得了一种平衡：它在平坦区域强制空间一致性，同时保留适应图像结构的能力，从而产生一个既可解释又能有效实现高质量去雾的权重网格。

#### B. 流形自适应采样分析

传统的高维插值方法（如三线插值）对每个像素应用固定的各向同性核，而不考虑局部图像结构。尽管计算效率高，但这种策略不可避免地混合了强边缘两侧的信息，导致边缘模糊，

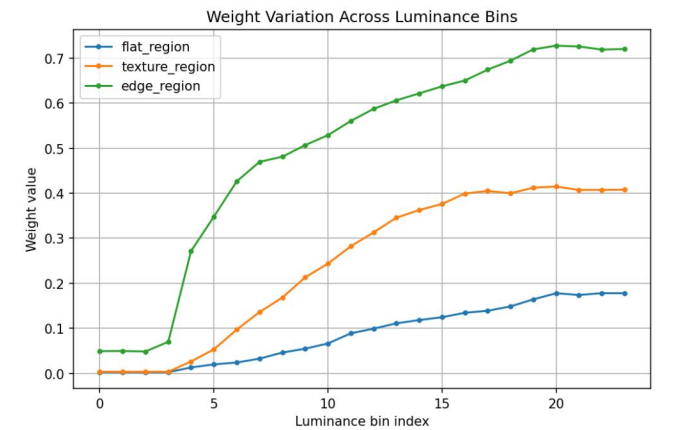


图7. 三个代表性区域在不同亮度分箱下的权重值。曲线分别对应平坦区域、纹理区域和边缘区域。边缘区域在所有亮度级别上始终保持较高的权重，以保留结构细节；而平坦区域整体权重较低，表明基于LUT的颜色映射已经足够。纹理区域则表现出中等权重并伴有轻微变化，说明学习到的权重网格能够适应局部图像内容。

光晕和颜色渗透——这些伪影在遥感影像中尤其有害，因为遥感影像必须保留精细的结构细节（如建筑物轮廓、道路网络）。为克服这一局限，我们的方法用流形自适应采样机制取代固定核，该机制尊重自然图像的内在低维结构。

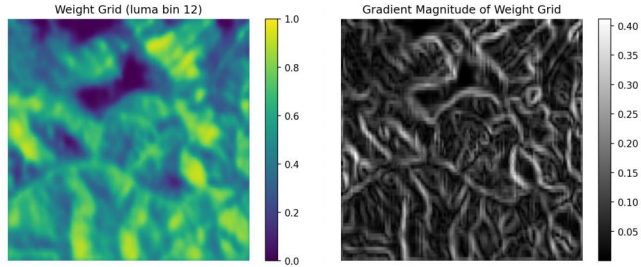


Fig. 8. **Smoothness analysis of the bilateral weight grid.** Left: weight map for the central luminance bin (viridis colormap). Right: gradient magnitude of the same weight map. The gradient is nearly zero in homogeneous areas, confirming that the edge-aware TV regularization effectively suppresses spurious noise and blocky artifacts. Strong gradients only appear along actual image edges, allowing the weight grid to change sharply where necessary to preserve fine structures.

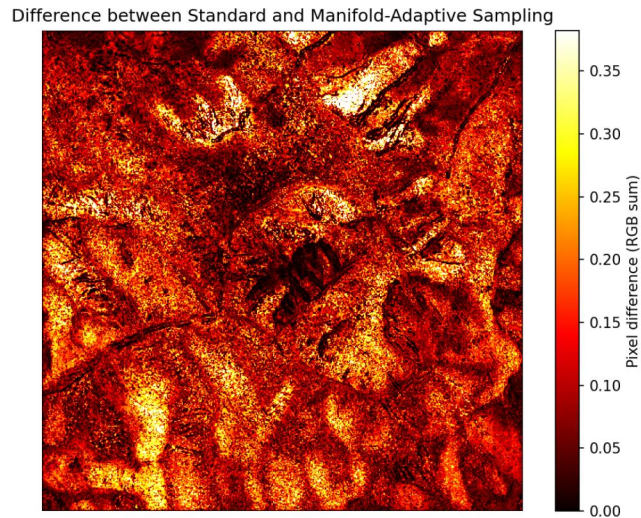


Fig. 9. **Difference map between standard and manifold-adaptive sampling.** The hot-colored map visualizes the per-pixel absolute difference (sum over RGB channels) between the outputs of standard trilinear interpolation and our manifold-adaptive sampling. Bright regions are concentrated along strong edges and textured areas, indicating that the adaptive mechanism primarily modifies the blending behavior where cross-edge mixing would otherwise occur, while leaving homogeneous regions nearly unchanged. This confirms that the manifold-adaptive sampling introduces targeted improvements without disturbing global color rendition.

images. Specifically, it estimates local edge orientation and strength from the luminance map, then dynamically adjusts the interpolation kernel: the kernel elongates along the edge direction to maintain continuity and suppresses weights across the edge to avoid mixing pixels from different objects. This design enables a single forward pass to simultaneously achieve global color stylization and local edge sharpening.

Fig. 9 visualizes the spatial distribution of differences between the outputs of standard trilinear interpolation and our manifold-adaptive sampling. Bright regions in the hot-colored map correspond to locations where the two sampling strategies diverge most. Notably, these bright regions are predominantly concentrated along strong edges and fine-textured areas, whereas homogeneous regions (e.g., sky, calm water) exhibit nearly zero difference. This observation confirms that the adaptive mechanism introduces targeted modifications only where cross-edge mixing would otherwise occur, leaving the

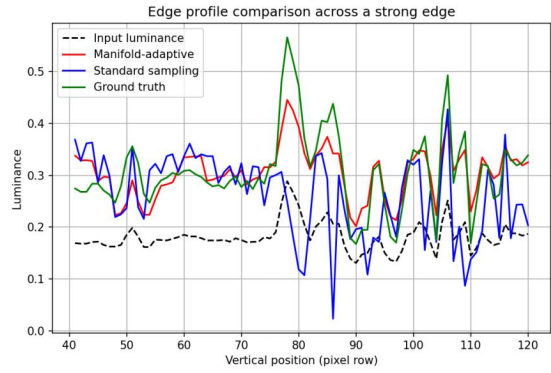


Fig. 10. **Edge profile comparison across a strong edge.** The plot shows luminance values along a vertical scanline crossing a salient edge. Compared with standard trilinear interpolation (blue), the proposed manifold-adaptive sampling (red) produces a substantially steeper transition that closely follows the ground truth (green), effectively suppressing cross-edge blur and preserving edge sharpness. The input luminance profile (black dashed) is also shown for reference.

global color mapping in flat areas unchanged. Consequently, the model improves edge fidelity without compromising the color consistency achieved by the LUT-based component.

Fig. 10 provides a quantitative comparison of luminance profiles along a vertical scanline that traverses a strong edge (e.g., a building-sky boundary). The input luminance (black dashed) shows an abrupt transition, characteristic of a sharp edge. Standard trilinear interpolation (blue) produces a much shallower slope, effectively blurring the edge and creating a gradual ramp that spreads across several pixels—a direct manifestation of cross-edge mixing. In contrast, our manifold-adaptive sampling (red) yields a substantially steeper transition that closely follows the input profile. The preservation of edge steepness directly translates into sharper boundaries and reduced halos in the final dehazed image.

In summary, the manifold-adaptive sampling mechanism addresses the fundamental shortcoming of fixed-kernel interpolation by explicitly respecting image edges. As evidenced by the difference map (Fig. 9) and the edge profile (Fig. 10), it effectively suppresses cross-edge blur while maintaining the ability to perform joint color rendition and detail enhancement in a single inference. The computational overhead remains negligible ( $\approx 0.05$  ms per 1080p image), making the mechanism both effective and practical for real-time deployment on edge devices.

## VI. CONCLUSION AND FUTURE WORKS

In this paper, we propose 6<sup>th</sup>Grid-Net, a unified remote sensing image dehazing framework that integrates LUT-based color rendition and bilateral grid-based edge preservation into a compact six-dimensional fusion tensor. By replacing conventional trilinear interpolation with a manifold-adaptive high-dimensional sampling mechanism that respects local edge orientation, texture strength, and color similarity, our method simultaneously achieves global color stylization and local detail enhancement in a single forward pass, avoiding the computational redundancy and feature loss of sequential pipelines. The introduced edge-aware grid smoothing regularization and

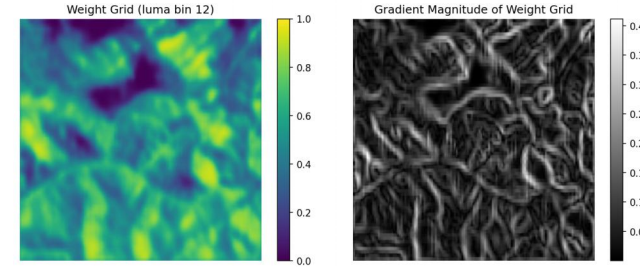


图 8. **双边权重网格的平滑性分析。**左：中央亮度分箱的权重图（viridis 颜色映射）。右：同一权重图的梯度幅值。在均匀区域中梯度几乎为零，证实边缘感知总变差正则化有效抑制了杂散噪声和块状伪影。强梯度仅沿实际图像边缘出现，使权重网格能够在必要处急剧变化以保留精细结构。

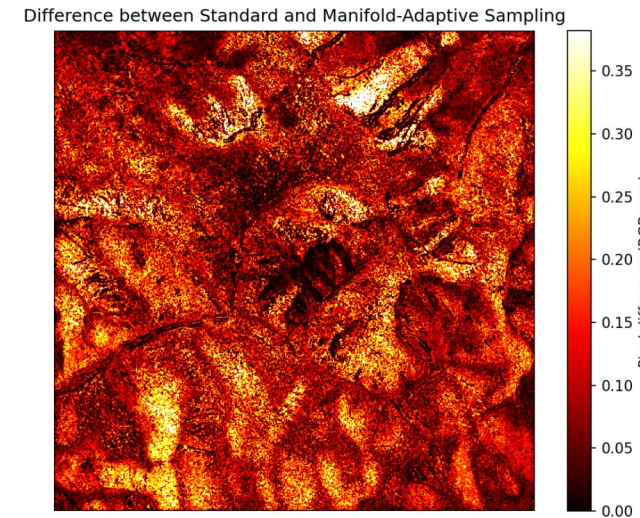


图 9. **标准采样与流形自适应采样之间的差异图。**热色图可视化了标准三线性插值与流形自适应采样输出之间的逐像素绝对差（RGB 通道之和）。高亮区域集中在强边缘和纹理区域，表明自适应机制主要修改了原本会发生跨边缘混合处的混合行为，同时使均匀区域几乎保持不变。这证实流形自适应采样在不干扰全局色彩还原的情况下引入了有针对性的改进。

具体而言，它从亮度图中估计局部边缘方向和强度，然后动态调整插值核：核沿边缘方向拉长以保持连续性，并抑制跨边缘的权重，避免混合来自不同物体的像素。这种设计使单次前向传播即可同时实现全局色彩风格化和局部边缘锐化。

图9 可视化了标准三线性插值与我们的流形自适应采样输出之间的空间差异分布。热色图中明亮区域对应两种采样策略差异最大的位置。值得注意的是，这些明亮区域主要集中在强边缘和细纹理区域，而均匀区域（如天空、平静水面）几乎显示为零差异。这一观察结果证实，自适应机制仅在原本会发生跨边缘混合的位置引入有针对性的修改，而保持

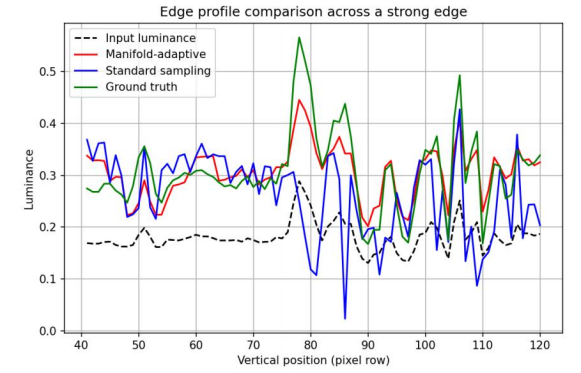


图10. **跨强边缘的边缘轮廓比较。**该图展示了沿一条穿过显著边缘的垂直扫描线的亮度值。与标准三线性插值（蓝色）相比，所提出的流形自适应采样（红色）产生了明显更陡峭的过渡，紧密跟随真实值（绿色），有效抑制了跨边缘模糊并保持了边缘锐度。输入亮度轮廓（黑色虚线）也一并显示以供参考。

平坦区域中的全局色彩映射不变。因此，该模型在不影响基于 LUT 的分量所实现的颜色一致性的前提下，提高了边缘保真度。

图10 对沿一条穿过强边缘（例如，建筑-天空边界）的垂直扫描线的亮度轮廓进行了定量比较。输入亮度（黑色虚线）显示出突变过渡，这是锐利边缘的特征。标准三线性插值（蓝色）产生了平缓得多的斜率，实际上使边缘模糊，并形成了一个横跨多个像素的渐变坡道——这是跨边缘混合的直接表现。相比之下，我们的流形自适应采样（红色）产生了明显更陡峭的过渡，紧密跟随输入轮廓。边缘陡峭度的保持直接转化为最终去雾图像中更锐利的边界和更少的光晕。

总之，流形自适应采样机制解决了固定核插值的根本缺陷，即显式地尊重图像边缘。如差异图（图9）和边缘轮廓（图10）所示，它有效抑制了跨边缘模糊，同时保持了在单次推理中联合执行色彩还原和细节增强的能力。计算开销仍然可以忽略不计（ $\approx 0.05$  毫秒/1080p 图像），使得该机制在边缘设备上实时部署时既有效又实用。

## VI. 结论与未来工作

本文提出 6<sup>th</sup>Grid-Net，一个统一的遥感图像去雾框架，将基于 LUT 的色彩还原与基于双边网格的边缘保持整合到一个紧凑的六维融合张量中。通过以流形自适应高维采样机制取代传统的三线性插值，该机制尊重局部边缘方向、纹理强度和颜色相似性，我们的方法在单次前向传递中同时实现全局色彩风格化和局部细节增强，避免了顺序流水线的计算冗余和特征损失。引入的边缘感知网格平滑正则化和

dynamic post-training quantization further reduce model size by approximately  $3\times$  without sacrificing reconstruction accuracy. Extensive experiments on the unified remote sensing dataset (SateHaze1K and RICE) demonstrate that 6<sup>th</sup>Grid-Net outperforms representative methods in terms of PSNR, SSIM, and LPIPS, while also exhibiting strong generalization to grayscale-to-RGB colorization on the ViCoW dataset. Moreover, the quantized model runs in real time on consumer-grade mobile devices, confirming its suitability for resource-constrained edge platforms. It should be noted, however, that the current six-dimensional tensor construction and manifold-adaptive sampling, though efficient, still involve a moderate memory footprint when scaling to ultra-high-resolution (e.g., 4K) imagery. Future work will explore adaptive grid pruning and learnable downsampling strategies to further reduce storage and computational costs while preserving edge fidelity and color accuracy.

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动态训练后量化进一步将模型大小减少约  $3\times$ ，同时不牺牲重建精度。在统一遥感数据集（SateHaze1K和RICE）上的大量实验表明，6<sup>th</sup>Grid-Net在PSNR、SSIM和LPIPS方面优于代表性方法，并且在ViCoW数据集上对灰度到RGB着色也表现出很强的泛化能力。此外，量化模型可在消费级移动设备上实时运行，证实了其适用于资源受限的边缘平台。但需要注意的是，当前的六维张量构建和流形自适应采样虽然高效，但在扩展到超高分辨率（如4K）图像时仍会产生中等程度的内存占用。未来工作将探索自适应网格剪枝和可学习下采样策略，以进一步降低存储和计算成本，同时保持边缘保真度和颜色精度。

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